

Alyson Gabbard Wilson, Ph.D.
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RESEARCH IMPACT

Alyson Wilson develops statistical methods for determining whether complex, high-consequence engineered systems will perform when they cannot be fully tested — the central problem in certifying systems too costly, dangerous, or rare to test to failure. Her core contribution is a body of Bayesian methodology for combining heterogeneous evidence — physical tests, computer simulations, component data, and expert judgment — into rigorous, quantified assessments of whole-system reliability. These methods reshaped how the reliability of the U.S. nuclear stockpile is assessed after the end of underground testing and have been applied across defense systems and in fields from materials science to aircraft-noise certification. She is a co-author of *Bayesian Reliability* and a Fellow of the American Statistical Association and the American Association for the Advancement of Science. She has been selected by the national laboratories to review their own engineering science — chairing a Los Alamos capability review and serving a decade on Sandia's Predictive Engineering Science panel.

EDUCATION

Ph.D. Statistics, Duke University, Durham, NC, May 1995

M.S. Statistics, Carnegie Mellon University, Pittsburgh, PA, August 1990

B.A. Mathematical Sciences, *summa cum laude*, Rice University, Houston, TX, May 1989

PROFESSIONAL EXPERIENCE

William & Mary, Williamsburg, VA

2025-present

Vice Provost for Research

2025-present

Lead the university's research enterprise and strategy as it establishes itself among the nation's Carnegie R1 institutions — building research infrastructure, development capacity, and the business processes supporting a growing research portfolio.

Professor, Department of Data Science

2025-present

North Carolina State University, Raleigh, NC

2013-2025

Interim Vice Chancellor for Research and Innovation

2024-2025

Directed the university's \$700M research enterprise — administration, compliance, intellectual property, and strategic partnerships — through a period of federal policy change.

***Senior Associate Vice Chancellor for Research Initiatives and Commercialization
Office of Research and Innovation***

2024-2025

Directed the national-security, quantum, and commercialization units (36 staff) and led three task forces restructuring the university's research administration — sponsored programs, research support services, and cross-boundary collaboration.

Associate Vice Chancellor for National Security and Special Research Initiatives 2020-2024
Office of Research and Innovation

Expanded the university's national-security research portfolio across defense, cybersecurity, and data science.

- Established master research agreements with major defense organizations, creating the foundation for multiple collaborative programs.
- Founded NC State's Data Science and AI Academy, building an interdisciplinary institution serving students and researchers across every college.
- Led the university-wide Data Science Initiative — mapping capabilities, convening faculty across colleges, and building the cross-college community that became the first of NC State's signature academies.

Principal Investigator, Laboratory for Analytic Sciences 2014-2024

As Principal Investigator, led the Laboratory for Analytic Sciences — a \$94M collaboration between the intelligence community and academia developing analytic methods and technologies for national-security problems — directing its research agenda, operations, compliance, and partnerships.

- Built a research program spanning 81 partners across four countries, delivering methods and tools for operational intelligence and security challenges.
- Established one of NC State's first large interdisciplinary research initiatives, creating the model and infrastructure for subsequent university-wide programs.

Professor, Department of Statistics 2015-2025

Associate Professor, Department of Statistics 2013-2015

Institute for Defense Analyses Science and Technology Policy Institute 2011-2019
Washington, DC

IDA Faculty Fellow 2015-2019

Adjunct Research Staff Member 2013-2019

Research Staff Member, Science and Technology Policy Institute and Systems
and Analyses Center/System Evaluation Division 2011-2013

Led analyses across defense and science-policy problems: reliability-assessment methods for developmental test and evaluation, dynamic management of chemical-biological-radiological-nuclear-explosives (CBRNE) detection assets, logistics requirements for Arctic research, and quantitative evaluation of federal science programs. Work informed DoD test and evaluation practice and national science-policy decisions.

Department of Statistics, Iowa State University, Ames, IA 2008-2013

Collaborating Associate Professor 2011-2013

Associate Professor 2008-2011

Statistical Sciences Group, Los Alamos National Laboratory 1999-present
Los Alamos, NM

Guest Scientist 2008-present

Technical Staff Member/Scientist 5 1999-2008

Developed and led the reliability-assessment methodology for conventional and nuclear weapon systems used in national stockpile stewardship after the end of underground nuclear testing — the problem of certifying whole-system reliability from surveillance data, physics, and simulation rather than live tests.

- Served as principal investigator and technical lead for the Joint DoD/DOE Munitions Program Stockpile Reliability Assessment, and as Systems Main Technical Element lead for the DOE Enhanced Surveillance Campaign.
- Created methods, adopted into stockpile surveillance practice, for combining heterogeneous evidence — component tests, computer models, and expert judgment — into quantified system-reliability assessments used in annual weapon assessments and surveillance decisions.
- Recognized with multiple DOE Defense Programs Awards of Excellence and Los Alamos Achievement and Director's Distinguished Performance Awards.

Cowboy Programming Resources, Inc., El Paso, TX	1995-1999
<i>Statistician/Senior Operations Research Systems Analyst</i>	1997-1999
<i>Statistician/Operations Research Systems Analyst</i>	1995-1996

Supported U.S. Army operational test and evaluation of air-defense and weapon systems — designing test data collection, sample-size and analysis plans, and developing original methodology for evaluating system performance, including PATRIOT PAC-3, the Joint Tactical Ground Station (JTAGS), and the Bradley Stinger Fighting Vehicle.

Laboratory of Statistical and Mathematical Methodology, Division of Computer Research and Technology, National Institutes of Health, Bethesda, MD	1991-1992
<i>Mathematical Statistician (GS-1529-9/11)</i>	1991-1992

Statistical methodology and computing consultant supporting NIH research-computing users and collaborating with biomedical researchers on clinical and analytic projects.

REFEREED PUBLICATIONS

150+ peer-reviewed publications, panel reports, and technical reports, including 3 books. Primary research focus is reliability assessment for complex systems, focusing on Bayesian methods and uncertainty quantification for heterogeneous information. Other areas include defense, intelligence, and counterterrorism; engineering and industrial applications; materials science.

Selected Publications

- M. S. Hamada, A. G. Wilson, C. S. Reese, H. F. Martz (2008). *Bayesian Reliability*. Springer, New York.
- A. Wilson, T. Graves, M. Hamada, C. S. Reese (2006). *Advances in Data Combination, Analysis, and Collection for System Reliability Assessment*. *Statistical Science* 21(4): 514-531.
- J. Guo, A. Wilson (2013). *Bayesian Methods for Estimating the Reliability of Complex Systems using Heterogeneous Multilevel Information*. *Technometrics* 55(4): 461-472.
- C. S. Reese, A. Wilson, J. Guo, M. Hamada, V. Johnson (2011). *A Bayesian Model for Integrating Multiple Sources of Lifetime Information in System-Reliability Assessments*. *Journal of Quality Technology* 43(2): 127-141.
- A. Wilson, A. Huzurbazar (2007). *Bayesian Networks for Multilevel System Reliability*. *Reliability Engineering and Systems Safety* 92(10): 1413-1420.
- X. Zhang, A. Wilson (2017). *System Reliability and Component Importance under Dependence: A Copula Approach*. *Technometrics* 59(2): 215-224.
- A. Wilson, K. Fronczyk (2017). *Bayesian Reliability: Combining Information*. *Quality Engineering* 29(1): 119-129.
- C. Anderson-Cook, T. Graves, M. Hamada, N. Hengartner, V. Johnson, C. S. Reese, A. Wilson (2007). *Bayesian Stockpile Reliability Methodology for Complex Systems with Application to a Munitions Stockpile*. *Journal of the Military Operations Research Society* 12(2): 25-38.
- R. Dickinson, L. Freeman, B. Simpson, A. Wilson (2015). *Statistical Models for Combining Information: Stryker Reliability Case Study*. *Journal of Quality Technology* 47(4): 400-415.
- C. Fancher, Z. Han, I. Levin, K. Page, B. Reich, R. Smith, A. Wilson, J. Jones (2016). *Use of Bayesian Inference in Crystallographic Structure Refinement via Full Diffraction Profile Analysis*. *Scientific Reports* 6:31625.

Books

- M. S. Hamada, A. G. Wilson, C. S. Reese, H. F. Martz (2008). *Bayesian Reliability*. Springer, New York. Chinese edition (2013).
- A. Wilson, G. Wilson, D. Olwell, eds (2006). *Statistical Methods in Counterterrorism: Game Theory, Modeling, Syndromic Surveillance, and Biometric Authentication*. Springer, New York.
- A. Wilson, N. Limnios, S. Keller-McNulty, Y. Armijo, eds (2005). *Modern Statistical and Mathematical Methods in Reliability*. World Scientific, Singapore. Chinese edition (2016).

Patents and Deployed Software

- L. Wendelberger, J. Gray, B. Reich, A. Wilson, S. Holloway (2022). roboBayes: Robust Online Bayesian Monitoring, <https://cran.r-project.org/web/packages/roboBayes/index.html>.
- E. Chaney, D. Fritsch, S. Pizer, V. Johnson, A. Wilson (1999). U.S. Patent 5926568. *Image Object Matching using Core Analysis and Deformable Shape Loci*. Licensed by Accuray (2013).

Journal Articles

- L. Wendelberger, B. Reich, A. Wilson, J. Gray (2026). [Detecting Deforestation Using Robust Online Bayesian Monitoring](#). *Data Science in Science* 5(1).
- C. Edwards, R. Smith, J. Mattingly, A. Wilson (2025). [Localization of Stationary and Moving Radiation Sources Using a Feedforward Neural Network with an Array of Sensors](#). *Nuclear Technology* 211(11): 2832-2845.
- H. Kim, A. Wilson (2025). [Comparing Risks for Binomial Reliability Assurance Test Planning](#). *Applied Stochastic Models in Business and Industry*, 41(3): e2912.
- A. Hollis, N. Clark, T. Moore, A. Wilson (2024). [From FMECA to Decision: A Fully Bayesian Reliability Process](#). *Military Operations Research Journal* 29(1): 45-62.
- L. Wendelberger, J. Gray, A. Wilson, R. Houborg, B. Reich (2022). [Multiresolution Broad Area Search: Monitoring Spatial Characteristics of Gapless Remote Sensing Data](#). *Journal of Data Science* 20(4): 545-565.
- J. Cahoon, K. Sanborn, A. Wilson (2021). [Practical Reliability Growth Modeling](#). *Quality and Reliability Engineering International* 37(7): 3108-3124.
- J. Bakerman, K. Pazdernik, G. Korkmaz, A. Wilson (2021). [Dynamic Logistic Regression and Variable Selection: Forecasting and Contextualizing Civil Unrest](#). *International Journal of Forecasting* 38(2): 648-661.
- A. Hollis, R. Smith, A. Wilson (2021). [Surrogate-Based Mutual Information Approximation and Optimization for Urban Source Localization](#). *International Journal for Uncertainty Quantification* 11(5): 39-55.
- R. Durodoye, M. Gumpertz, A. Wilson, E. Griffith, S. Ahmad (2020). [Tenure and Promotion Outcomes at Four Large Land Grant Universities: Examining the Role of Gender, Race, and Academic Discipline](#). *Research in Higher Education* 61(5): 628-651.
- J. Lee, J. Rathsam, A. Wilson (2020). [Bayesian Statistical Models for Community Annoyance Survey Data](#). *Journal of the Acoustical Society of America* 147(4): 2222-2234.
- A. Chakraborty, S. Lahiri, A. Wilson (2020). [A Statistical Analysis of Noisy Crowdsourced Weather Data](#). *Annals of Applied Statistics* 14(1): 116-142.
- R. Broughton, S. O'Donnell, E. Gabilondo, C. Chung, P. Maggard, A. Wilson, B. Reich, R. Smith, J. Jones (2020). [Bayesian Refinement of Full Profile Diffraction Patterns for](#)

Uncertainty Quantification. *Acta Crystallographica, Section A: Foundations and Advances* 76(a1).

- J. Gilman, K. Fronczyk, A. Wilson (2019). Bayesian Modeling and Test Planning for Multi-phase Reliability Assessment. *Quality and Reliability Engineering International* 35(3): 750-760.
- A. Wilson, M. Schmidt, L. Schmidt, B. Winter (2019). Immersive Collaboration on Data Science for Intelligence Analysis: An Interview with Alyson Wilson and Matthew Schmidt by Lara Schmidt and Brent Winter. *Harvard Data Science Review* 1(2).
- E. Typhina, A. Wilson (2019). Discussion of “Effective Interdisciplinary Collaboration Between Statisticians and Other Subject Matter Experts.” *Quality Engineering* 31(1): 192-194.
- Y. Tian, H. Bondell, A. Wilson (2019). Bayesian Variable Selection for Logistic Regression. *Statistical Analysis and Data Mining* 12(5): 378-393.
- J. Jones, R. Broughton, T. Iamsasri, C. Fancher, A. Wilson, B. Reich, R. Smith (2019). The Use of Bayesian Inference in the Characterization of Materials and Thin Films. *Acta Crystallographica, Section A: Foundations and Advances* 75(a209).
- K. Gasior, N. Wagner, J. Cores, R. Caspar, A. Wilson, S. Bhattacharya, M. L. Hauck (2019). The Influence of Cellular Contact and the TGF- β Signaling Pathway on the Activation of the Epithelial Mesenchymal Transition. *Cell Adhesion and Migration*, 13(1): 63-75.
- H. Rendon, A. Wilson, J. Stegall (2018). Is it “Fake News”? Intelligence Community Expertise and News Dissemination as Measurement for Media Reliability. *Intelligence and National Security* 33(7): 1040-1052.
- J. Bakerman, K. Pazdernik, A. Wilson, R. Bahrn, G. Fairchild (2018). Twitter Geolocation: A Hybrid Approach. *ACM Transactions on Knowledge Discovery from Data* 20(3): 1-17.
- R. Zoh, A. Wilson, S. Vander Wiel, E. Lawrence (2018). Using the Negative Log-Gamma Distribution for System Reliability Assessment. *Journal of Risk and Reliability* 232(3): 308-319.
- B. Weaver, M. Hamada, A. Wilson, J. Bakerman (2017). Bayesian Assurance Tests for Degradation Data. *Quality and Reliability Engineering International* 33(8): 2699-2709.
- X. Zhang, A. Wilson (2017). System Reliability and Component Importance under Dependence: A Copula Approach. *Technometrics* 59(2): 215-224.
- A. Wilson, K. Fronczyk (2017). Bayesian Reliability: Combining Information. *Quality Engineering* 29(1): 119-129. (*This is one of the top 10 most read articles in the history of the journal.*)
- M. Gumpertz, R. Durodoye, E. Griffith, A. Wilson (2017). Retention and Promotion of Women and Minority Faculty in Science and Engineering. *PLoS ONE*, 12(11): e0187285.
- T. Iamsasri, J. Guerrier, G. Esteves, C. Fancher, A. Wilson, R. Smith, E. Paisley, R. Johnson-Wilke, J. Ihlefeld, N. Bassiri-Gharb, J. Jones (2017). A Bayesian Approach to Modeling Diffraction Profiles and Application to Ferroelectric Materials. *Journal of Applied Crystallography* 50: 211-220.
- K. Gasior, M. Hauck, A. Wilson, S. Bhattacharya (2017). A Theoretical Model of the Wnt Signaling Pathway in the Epithelial Mesenchymal Transition (EMT). *Theoretical Biology and Medical Modelling* 14:19.
- L. Dewald, R. Holcomb, S. Parry, A. Wilson (2016). A Bayesian Approach to Evaluation of Operational Testing of Land Warfare Systems. *Journal of the Military Operations Research Society* 21(4): 23-32.

- C. Fancher, Z. Han, I. Levin, K. Page, B. Reich, R. Smith, A. Wilson, J. Jones (2016). Use of Bayesian Inference in Crystallographic Structure Refinement via Full Diffraction Profile Analysis. *Scientific Reports* 6:31625.
- H. Graham, A. Motsinger-Reif, J. Buse, T. Havener, A. Wilson, M. Wagner, D. Rotroff, ACCORD/ACCORDion Investigators (2016). Incorporating Concomitant Medications into Genome-Wide Analyses for the Study of Complex Disease and Drug Response. *Frontiers in Genetics/Applied Genetic Epidemiology* 7:138.
- R. Brost, C. Phillips, D. Robinson, D. Stracuzzi, A. Wilson, D. Woodbridge (2015). Computing Quality Scores and Uncertainty for Approximate Pattern Matching in Geospatial Semantic Graphs. *Statistical Analysis and Data Mining* 8(5/6): 340-352.
- D. Nordman, J. Berry, C. Phillips, L. Fostvedt, A. Wilson, C. Seshadhri (2015). Why Do Simple Algorithms for Triangle Enumeration Work in the Real World? *Internet Mathematics* 11(6): 555-571.
- R. Dickinson, L. Freeman, B. Simpson, A. Wilson (2015). Statistical Models for Combining Information: Stryker Reliability Case Study. *Journal of Quality Technology* 47(4): 400-415.
- C. S. Reese, A. Wilson (2014). Discussion of "Methods for Planning Repeated Measures Accelerated Degradation Tests." *Applied Stochastic Models in Business and Industry* 30(6): 674-676.
- M. Hamada, A. Wilson, B. Weaver, R. Griffiths, H. Martz (2014). Bayesian Binomial Assurance Tests for System Reliability Using Component Data. *Journal of Quality Technology* 46(1): 24-32. *Paper selected for JQT invited session at 2014 INFORMS Annual Meeting.*
- J. Wendelberger, A. Wilson, S. Stinnett, B. Gaydos (2014). Working in Interdisciplinary Teams. *Chance* 27(4): 31-33.
- E. Casleton, U. Genschel, A. Wilson (2014). A Pilot Study for Teaching Metrology in an Introductory Statistics Course. *Journal of Statistics Education* 22(3).
- J. Guo, A. Wilson (2013). Bayesian Methods for Estimating the Reliability of Complex Systems using Heterogeneous Multilevel Information. *Technometrics* 55(4): 461-472.
- J. Guo, D. Nordman, A. Wilson (2013). Bayesian Nonparametric Models for Community Detection. *Technometrics* 55(4): 390-402.
- B. Weaver, M. Hamada, S. Vardeman, A. Wilson (2012). A Bayesian Approach to the Analysis of Gauge R&R Data. *Quality Engineering* 24(4): 486-500.
- C. M. Anderson-Cook, L. Lu, G. Clark, S. P DeHart, R. Hoerl, B. Jones, R. J. MacKay, D. C. Montgomery, P. A. Parker, J. Simpson, R. Snee, S. Steiner, J. Van Mullekom, G. G. Vining, A. G. Wilson (2012). Statistical Engineering—Forming the Foundations. *Quality Engineering* 24: 110-132.
- C. M. Anderson-Cook, L. Lu, G. Clark, S. P DeHart, R. Hoerl, B. Jones, R. J. MacKay, D. C. Montgomery, P. A. Parker, J. Simpson, R. Snee, S. Steiner, J. Van Mullekom, G. G. Vining, A. G. Wilson (2012). Statistical Engineering—Roles for Statisticians and the Path Forward. *Quality Engineering* 24: 133-152.
- L. Lu, C. Anderson-Cook, A. Wilson (2011). Choosing a Consumption Strategy for a Population of Systems based on Reliability. *Journal of Risk and Reliability (Proceedings of the Institution of Mechanical Engineers, Part O)*, 225(4): 407-423.
- A. Wilson, C. Anderson-Cook, A. Huzurbazar (2011). A Case Study for Quantifying System Reliability and Uncertainty. *Reliability Engineering and System Safety*, 96(9): 1076–1084.

- C. Anderson-Cook, S. Crowder, A. Huzurbazar, J. Lorio, J. Ringland, A. Wilson (2011). Quantifying Reliability Uncertainty from Catastrophic and Margin Defects: A Proof of Concept. *Reliability Engineering and System Safety* 96(9): 1063-1075.
- C. S. Reese, A. Wilson, J. Guo, M. Hamada, V. Johnson (2011). A Bayesian Model for Integrating Multiple Sources of Lifetime Information in System-Reliability Assessments. *Journal of Quality Technology* 43(2): 127-141.
- S. Vander Wiel, A. Wilson, T. Graves, C. S. Reese (2011). A Random Onset Model for Degradation of High-Reliability Systems. *Technometrics* 53(2): 163-172.
- M. Hamada, A. Huzurbazar, S. Vander Wiel, A. Wilson (2011). Assessing the Risks of Sampling Rates for Surveilling a Population. *Quality Engineering* 23(3): 242-252.
- A. Wilson, C. Anderson-Cook (2010). Discussion of "Reliability Growth Management Metrics and Statistical Methods for Discrete-Use Systems." *Technometrics* 52(4): 397-400.
- A. Wilson, A. Huzurbazar, K. Sentz (2009). The Imprecise Dirichlet Model For Multilevel Systems Reliability. *Journal of Statistical Theory and Practice* 3(1): 211-223.
- G. S. Parnell, L. L. Borio, L. A. Cox, G. G. Brown, S. Pollock, A. G. Wilson (2009). Response to Ezell and von Winterfeldt. *Biosecurity and Bioterrorism: Biodefense Strategy, Practice, and Science* 7(1): 111-112.
- N. Singpurwalla, A. Wilson (2008). Probability, Chance, and the Probability of Chance. *IIE (Institute for Industrial Engineers) Transactions* 41(1): 12-22.
- C. Anderson-Cook, T. Graves, N. Hengartner, R. Klamann, A. Koehler, A. Wilson, G. Anderson, G. Lopez (2008). Reliability Modeling Using Both System Test and Quality Assurance Data. *Journal of the Military Operations Research Society* 13: 5-18.
- G. Parnell, L. Borio, D. Banks, G. Brown, A. Wilson (2008). Scientists Urge DHS to Improve Bioterrorism Risk Assessment. *Biosecurity and Bioterrorism: Biodefense Strategy, Practice, and Science* 6(4): 1-4.
- A. Wilson, A. Huzurbazar (2007). Bayesian Networks for Multilevel System Reliability. *Reliability Engineering and Systems Safety* 92(10): 1413-1420.
- A. Wilson, L. McNamara, G. Wilson (2007). Information Integration for Complex Systems. *Reliability Engineering and Systems Safety* 92(1): 121-130.
- C. Anderson-Cook, T. Graves, M. Hamada, N. Hengartner, V. Johnson, C. S. Reese, A. Wilson (2007). Bayesian Stockpile Reliability Methodology for Complex Systems with Application to a Munitions Stockpile. *Journal of the Military Operations Research Society* 12(2): 25-38.
- A. Wilson, T. Graves, M. Hamada, C. S. Reese (2006). Advances in Data Combination, Analysis, and Collection for System Reliability Assessment. *Statistical Science* 21(4): 514-531.
- S. Keller-McNulty, A. Wilson, C. Anderson-Cook (2006). Reliability. *Statistical Science* 21(4): 427.
- S. Keller-McNulty, G. Wilson, A. Wilson (2005). The Impact of Technology on the Scientific Method, with discussion. *Chance* 18(4): 4-8.
- M. Hamada, H. Martz, C. S. Reese, T. Graves, V. Johnson, A. Wilson (2004). A Fully Bayesian Approach for Combining Multilevel Failure Information in Fault Tree Quantification and Corresponding Optimal Resource Allocation. *Reliability Engineering and Systems Safety* 86(3): 297-305.
- A. Wilson, M. Hamada, M. Xu (2004). Assessing Production Quality with Nonstandard Measurement Errors. *Journal of Quality Technology* 36(2): 193-206.

- C. S. Reese, A. Wilson, M. Hamada, H. Martz, K. Ryan (2004). Integrated Analysis of Computer and Physical Experiments. *Technometrics* 46(2): 153-164.
- M. Hamada, H. Martz, C. S. Reese, A. Wilson (2001). Finding Near-Optimal Bayesian Experimental Designs via Genetic Algorithms. *The American Statistician* 55(3): 175-181.
- A. Wilson (1998). Sample Size Calculations for Test and Evaluation. *The ITEA Journal of Test and Evaluation* 19(September/October): 46-51.
- A. Knebel, S. Janson-Bjerklie, J. Malley, A. Wilson, J. Marini (1994). Comparison of Breathing Comfort During Weaning with Two Ventilatory Modes. *American Journal of Respiratory and Critical Care Medicine* 149(January): 14-18.

Book Chapters

- S. Singh, A. Paterson, L. Wendelberger, C. Fancher, B. Reich, R. Smith, A. Wilson, J. Jones (2020). Algorithms in Profile Diffraction Analysis. Handbook on Big Data and Machine Learning in the Physical Sciences Volume 1: Big Data Methods in Experimental Materials Discovery, eds. I. Foster and S. Kalinin, World Scientific, 501-539.
- A. Paterson, B. Reich, R. Smith, A. Wilson, J. Jones (2018). Bayesian Approaches to Uncertainty Quantification and Structure Refinement from X-ray Diffraction. Materials Discovery and Design: Data Science and Optimal Learning, ed. T. Lookman, Springer, 81-102.
- A. Wilson, K. Fronczyk (2017). National Security Risk Analysis. Wiley StatsRef: Statistics Reference Online 1, eds. N. Balakrishnan, T. Colton, B. Everitt, W. Piegorisch, F. Ruggeri, J. Teugels.
- A. Wilson (2007). Hierarchical Markov Chain Monte Carlo for Bayesian System Reliability. Encyclopedia of Statistics in Quality and Reliability, F. Ruggeri, R. Kenett, F. Faltin, eds. Wiley, Chichester, UK, 824-828.
- S. Keller-McNulty, A. Wilson. Reliability for the 21st Century (2003). In Mathematical and Statistical Methods in Reliability, B. Lindqvist, K. Doksum, eds., World Scientific, 15-30.
- A. Wilson, C. S. Reese, M. Hamada, and H. Martz (2003). Integrated Analysis of Computational and Physical Experimental Lifetime Data. In Mathematical Reliability: An Expository Perspective, R. Soyer, N. Singpurwalla, T. Mazzuchi, eds. Kluwer Press, 183-194.
- A. Wilson, V. Johnson. Models for Shape Deformation (1995). In Bayesian Statistics 5, J. Berger, J. Bernardo, A. P. Dawid, A. F. M. Smith, eds. Oxford University Press, 801-808.

Conference Publications

- K. Hameed, R. Johnston, B. Younce, M. Tang, A. Wilson. Motif-Based Exploratory Data Analysis for Malign Influence (2023). *Proceedings of the 2023 International AAAI Conference on Web and Social Media*.
- K. Pazdernik, B. Reich, K. Page, A. Wilson (2017). Hierarchical Bayesian Modeling of Atomic Structural Disorder. *Proceedings of the International Conference on Mathematics and Computational Methods Applied to Nuclear Science and Engineering*, Jeju, Korea.
- W. C. Lenhardt, M. Conway, E. Scott, B. Blanton, A. Krishnamoorthy, M. Hadzikadic, M. Vouk, A. Wilson (2016). Cross-Institutional Research Cyberinfrastructure for Data Intensive Science. *Proceedings of the IEEE High-Performance Extreme Computing Conference (HPEC) 2016*.

- J. Berry, L. Fostvedt, D. Nordman, C. Phillips, C. Seshadhri, A. Wilson (2014). Why Do Simple Algorithms for Triangle Enumeration Work in the Real World? *Proceedings of 5th Innovations in Theoretical Computer Science (ITCS) Conference*, 225-234.
- K. Sentz, A. Wilson (2005). Fault Tree Uncertainty Quantification using Probabilities and Belief Structures on Basic and Non-Basic Events. *NAFIPS 2005: 2005 Annual Meeting of the North American Fuzzy Information Processing Society*, Detroit, MI, 65-68.
- A. Wilson, V. Johnson, S. Pizer, D. Fritsch, L. Yu, E. Chaney (1996). Towards a Framework for Automated Image Analysis. *Proceedings of the Leeds Annual Statistical Workshop: Image Fusion and Shape Variability Techniques*, Leeds, UK, 13-20.
- A. Wilson, V. Johnson (1994). Priors on Scale-Space Templates. Proceedings Volume 2299, Mathematical Methods in Medical Imaging III, *International Society for Optical Engineering (SPIE)*, 161-168.

Peer-Reviewed Panel Reports

- Statistical and Data-Driven Methods for Additive Manufacturing Qualification: Proceedings of a Workshop. Committee on Statistical and Data-Driven Methods for Additive Manufacturing Qualification: Future Directions and Opportunities in Materials and Models, *National Academies of Sciences, Engineering, and Medicine*, 2024.
- Emerging Hazards in Commercial Aviation—Report 2: Ensuring Safety During Transformative Changes. Committee on Emerging Trends in Aviation Safety, *National Academies of Sciences, Engineering, and Medicine*, 2024.
- Risk Analysis Methods for Nuclear War and Nuclear Terrorism, Phase 2 Report and Risk Analysis Methods for Nuclear War and Nuclear Terrorism, Phase 2 Report (abbreviated). Committee on Risk Analysis Methods for Nuclear War and Nuclear Terrorism, *National Academies of Sciences, Engineering, and Medicine*, 2023.
- Risk Analysis Methods for Nuclear War and Nuclear Terrorism, Phase 1 Report. Committee on Risk Analysis Methods for Nuclear War and Nuclear Terrorism, *National Academies of Sciences, Engineering, and Medicine*, 2023.
- Emerging Hazards in Commercial Aviation—Report 1: Initial Assessment of Safety Data and Analysis Processes. Committee on Emerging Trends in Aviation Safety, *National Academies of Sciences, Engineering, and Medicine*, 2022.
- An Assessment of Selected Divisions of the Information Technology Laboratory at the National Institute of Standards and Technology. Committee on NIST Technical Programs, National Academies of Sciences, Engineering, and Medicine, 2021.
- Empowering the Defense Acquisition Workforce to Improve Mission Outcomes Using Data Science. Committee for Improving Defense Acquisition Workforce Capability in Data Use, National Academies of Sciences, Engineering, and Medicine, 2021.
- Improving Defense Acquisition Workforce Capability in Data Use: Proceedings of a Workshop in Brief. Committee for Improving Defense Acquisition Workforce Capability in Data Use, National Academies of Sciences, Engineering, and Medicine, 2020.
- Learning from the Science of Cognition and Perception for Decision Making: Proceedings of a Workshop. Steering Committee on Learning from the Science of Cognition and Perception for Decision Making supporting the Committee on a Decadal Survey of Social and Behavioral Sciences for Applications to National Security, National Academies of Sciences, Engineering, and Medicine, 2018.

Complex Decision Making in Networked Systems of Humans and Machines: A Multidisciplinary Approach. Committee on Integrating Humans, Machines and Networks: A Global Review of Data-to-Decision Technologies, National Research Council of the National Academies, 2014.

Assessing the Reliability of Complex Models: Mathematical and Statistical Foundations of Verification, Validation, and Uncertainty Quantification. Committee on Mathematical Foundations of Verification, Validation, and Uncertainty Quantification, National Research Council of the National Academies, 2012.

Testing of Body Armor Materials, Phase III. Committee to Review the Testing of Body Armor Materials for Use by the U.S. Army, National Research Council of the National Academies, 2012.

Industrial Methods for the Effective Testing and Development of Defense Systems. Panel on Industrial Methods for the Effective Test and Development of Defense Systems National Research Council of the National Academies, 2011.

Testing of Body Armor Materials for Use by the U.S. Army, Phase II. Committee to Review the Testing of Body Armor Materials for Use by the U.S. Army, National Research Council of the National Academies, 2010.

Phase I Report on Review of the Testing of Body Armor Materials for Use by the U.S. Army. Committee to Review the Testing of Body Armor Materials for Use by the U.S. Army, National Research Council of the National Academies, 2009.

Department of Homeland Security Bioterrorism Risk Assessment: A Call for Change. Committee on Methodological Improvements to the Department of Homeland Security's Biological Agent Risk Analysis, National Research Council of the National Academies, 2008.

Mathematics for Analysis of Petascale Data: Report on a Department of Energy Workshop. Applied Mathematics Program of the Office of Advanced Scientific Computing Research (ASCR), Office of Science, Department of Energy, 2008.

Improved Operational Testing and Evaluation and Methods of Combining Test Information for the Stryker Family of Vehicles and Related Army Systems: Phase II Report. Panel on Operational Test Design and Evaluation of the Interim Armored Vehicle, National Research Council of the National Academies, 2004.

Improved Operational Testing and Evaluation: Better Measurement and Test Design for the Interim Brigade Combat Team with Stryker Vehicles, Phase I Report. Panel on Operational Test Design and Evaluation of the Interim Armored Vehicle, National Research Council of the National Academies, 2003.

Test Design and Evaluation for the Interim Armored Vehicle. Panel on Operational Test Design and Evaluation of the Interim Armored Vehicle, National Research Council of the National Academies, 2002.

Peer-Reviewed Technical Reports

J. Lee, J. Rathsam, A. Wilson (2019). Statistical Modeling of Quiet Sonic Boom Community Response Survey Data. NASA/TM-2019-220427.

M. Avery, J. Cartier, S. Cazares, P. Dolph, J. Fregeau, J. Holzer, K. Morrison, S. Nunes, K. Papadantonakis, S. Renn, J. Snyder, K. Spencer, A. Wilson (2013). Review of Methods and Algorithms for Dynamic Management of CBRNE Collection Assets. IDA Document ID P-4995.

- M. Ambroso, A. Kelley, A. Wilson (2013). Reliability Basics: Key Reliability Concepts for DT&E. IDA Document ID P-4925.
- E. Adelizzi, R. Bontz, J. Fleury, A. Kelley, A. Kim, R. Mahoney, J. Palguta, S. Renn, G. Sharp, J. Urban, R. Uy, A. Wilson (2013). Analytical Support to the Hard and Deeply Buried Target Defeat Deep Dive Team: Final Report. IDA Document ID P-4998.
- S. Jonas and A. Wilson. Foresight and Understanding from Scientific Exposition (2012). IDA Document ID 2012-08482.
- B. Lal, S. Jonas, A. Wilson, E. Lee, A. Richards, V. Peña (2012). An Outcome Evaluation of the National Institutes of Health (NIH) Director's Pioneer Award (NDPA) Program, FY 2004-2006. IDA Document ID P-4899.
- B. Lal, S. Jonas, A. Wilson, E. Lee, A. Richards, V. Peña (2012). Preliminary Findings of Outcome Evaluation of the National Institutes of Health (NIH) Director's Pioneer Award (NDPA) Program, FY 2004-2006. IDA Document ID 2012-07132.
- M. Ambroso, B. Barrois, J. Buontempo, D. DeRiggi, J. Fox, M. Fries, K. Guerrero, J. Hong, D. Hunter, A. Kim, S. Ouellette, G. Sharp, R. Uy, G. Willmes, A. Wilson (2012). Independent Review and Assessment of the Ground-Based Midcourse Defense System. IDA Document ID P-4802.
- L. Freeman, A. Wilson, M. Nuñez (2012). Leveraging Simulation Rools into Operational Test Solutions for MQ-9 Using Experimental Design and Analysis Techniques (Memorandum). IDA Document ID 2012-06610.
- B. Lal, S. Jonas, A. Wilson, E. Lee, V. Peña, A. Richards (2012). NDPA Program Evaluation Study Design (Memorandum). IDA Document ID 2012-06285.
- P. Flattau, S. Nash, M. Nuñez, A. Wilson, D. Basco, A. Laskey, S. Jonas (2011). Overlap, Redundancy, and Fragmentation in the 2011 Inventory of Federal STEM Education. IDA Document ID 2011-06156.
- P. Kegelmeyer et al. (2010). Network Discovery, Characterization, and Prediction: A Grand Challenge LDRD Final Report. Sandia National Laboratories SAND Report 2010-8715.
- C. Anderson-Cook, M. Hamada, A. Huzurbazar, A. Wilson, Q. Fatherley (2009). B-61 Subsystem Resource Allocation Case Study. LA-UR-09-05932.
- D. Monroe, A. Huzurbazar, E. Kelly, A. Wilson, D. Ceman, J. Trujillo, D. Borovina (2009). An Element of Test Fire Optimization: Detonator Reliability for Potential Future Designs (Preliminary Report). LA-CP-09-00726.
- A. Wilson, Q. Fatherley, M. Mundt, M. Dvorack (2008). Case Study System Uncertainty Block Diagram. LA-UR-08-1000.
- R. Barber, B. Smith, R. Valicenti, S. McCready, K. Kelley, D. Crane, B. Aubert, F. Guerra, D. Harradine, S. Joyce, A. Wilson, G. Wilson, E. Lawrence, J. Reynolds, J. Bridgewater, M. Salmon, J. Mitchell, D. Schwartz, D. Brown, M. Prime, T. Sisneros, P. Burgardt, C. Necker, D. Veirs, T. Ickes, W. Ward, D. Gardiner, C. Cady, B. Taylor, R. Brunner, A. Chavez, D. Court, A. Sharif Heger, A. Charmatz, E. Schwegler, S. Abeln, D. Riha, B. Thacker (2007). Engineering Assessment of the Los Alamos National Laboratory—Manufactured Type 126 Pit. LA-CP-07-0778.
- D. Harradine, S. Joyce, A. Wilson, G. Wilson, E. Lawrence, J. Reynolds, J. Bridgewater (2007). Requalification of Plutonium Chemical Aspects of LANL Manufactured Pits. LA-CP-07-0777.
- D. Monroe, A. Munger, K. Laintz, A. Wilson, J. Kramer (2007). PETN Stabilization: Thermal Stress Effects of Four PETN Batches. LA-CP-07-0804.

- T. Zocco, L. Morales, D. Teter, S. Larson, T. Stephens, A. Wilson, M. Elliot (2007). Enhanced Surveillance Campaign Input to the LANL 2007 Annual Assessment. LA-CP-07-0306.
- C. Anderson-Cook, T. Graves, E. Lawrence, S. Vander Wiel, A. Wilson, G. Anderson, N. Delacruz, G. Lopez (2007). Stockpile Reliability Estimates for the TOW Missile. LA-CP-07-0998.
- C. Anderson-Cook, T. Graves, A. Koehler, E. Lawrence, S. Vander Wiel, A. Wilson (2005). Comparing Different System Reliability Estimation Methods. LA-UR-05-9401.
- A. Wilson (2005). Nuclear Explosives Package Reliability Assessment Methodology. LA-CP-05-0904.
- A. Wilson, T. Graves, N. Hengartner, R. Klamann, A. Koehler, G. Anderson, K. Deal, G. Lopez (2004). Development of an Integrated Stockpile Reliability Model for the TOW Missile. LA-CP-04-0644.
- A. Wilson, S. Keller-McNulty (2003). Complex Information Integration for Reliability and Performance Estimation for the BMD Family of Systems. LA-UR-03-5598.
- N. Hengartner, A. Wilson (2003). Bayesian Stockpile Reliability Methodology for Complex Systems with Application to the STINGER Missile Stockpile. LA-CP-03-0894.
- S. Eubank, C. Barrett, S. Michalak, D. Roberts, A. Wilson, G. Wilson (2002). Assessment of Responses to Smallpox Attack: Report to the Office of Homeland Security. LA-CP-02-0284.
- C. S. Reese, J. Morzinski, A. Wilson (2002). Assessing Stinger Stockpile Reliability Using Information Integration Technology. LA-CP-02-0164.
- A. Wilson, T. Graves, G. Hemphill (2002). Statistical Analysis Appendix for Process Qualification Plan 145. LA-CP-02-0104.
- J. Booker, S. Keller-McNulty, M. Meyer, C. S. Reese, A. Wilson (2001). Probabilistic Information Integration Technology. LA-UR-01-1060.
- T. Graves, A. Wilson (2001). Statistical Analysis of Part Homogeneity. LA-CP-01-0450.
- T. Graves, A. Wilson (2001). Statistical Analysis of Gallium Concentrations. LA-CP-01-0449.
- A. Wilson (2001). Statistical Analysis of DC Arc Data and Statistical Analysis of RFP Tbred Data. LA-CP-01-0346.
- T. Graves, A. Wilson (2001). Statistical Analysis of Plutonium Exchange Data (1985-1988) and AWE Exchange Data. LA-CP-01-0194.
- A. Wilson (2001). Chemical Analysis of WR Ingot Data. LA-CP-01-0193.
- T. Graves, A. Wilson (2001). Statistical Appendix for Process Qualification Plan 141a and Process Qualification Plan 150-Rev A: Plutonium Assay by Controlled Potential Coulometry and Dissolution. LA-CP-01-0187.
- T. Graves, A. Wilson (2001). Statistical Appendix for Process Qualification Plan 153-Rev A: Spectrophotometric Determination of Iron in Plutonium Materials. LA-CP-01-0186.
- T. Graves, A. Wilson (2001). Statistical Appendix for Process Qualification Plan 142-Rev A: Pu Isotopic Distributions of Pu Materials by Thermal Ionization Mass Spectrometry. LA-CP-01-0185.
- A. Wilson, S. Keller-McNulty (2000). Statistical Representations for Information Integration. LA-UR-00-4850.
- M. McNulty, A. Wilson, S. Keller-McNulty (2000). Information Integration Technology. LA-UR-00-4849.
- A. Wilson, M. Morris, J. Morrison (1999). NMD BMC2 Algorithm Analysis: Statistical Methods to Validate the National Missile Defense Battle Management Command and Control Software. LA-CP-99-0264.

Primary Author of *Draft Joint Tactical Ground Station Regression Testing System Assessment*. Prepared for U.S. Army Operational Evaluation Command, 1997.

Primary Author of *Draft Joint Tactical Ground Station Product Verification Test/First Article Test Operational Assessment*. Prepared for U.S. Army Operational Evaluation Command, 1996.

Contributing Author for *Draft Joint Tactical Ground Station Product Verification Test/First Article Test Independent Evaluation Briefing*. Prepared for U.S. Army Operational Evaluation Command, 1996.

Contributing Author for *Draft PATRIOT Advanced Capability-3, Configuration-2 Follow-on Test and Evaluation Analysis Report*. Prepared for U.S. Army Operational Evaluation Command, 1996.

Contributing Author for *Draft PATRIOT Advanced Capability-3, Configuration-2 Follow-on Test and Evaluation Test and Evaluation Report*. Prepared for U.S. Army Operational Evaluation Command, 1996.

Contributing Author for *Draft PATRIOT Advanced Capability-3, Configuration-2 Follow-on Test and Evaluation Independent Evaluation Briefing*. Prepared for U.S. Army Operational Evaluation Command, 1996.

Contributing Author for *Draft Joint Tactical Ground Station Developmental Test/Operational Test Analysis Report*. Prepared for U.S. Army Operational Evaluation Command, 1996.

Contributing Author for *Draft Joint Tactical Ground Station Developmental Test/Operational Test and Evaluation Report*. Prepared for U.S. Army Operational Evaluation Command, 1996.

Contributing Author for *Draft Joint Tactical Ground Station Developmental Test/Operational Test Independent Evaluation Briefing*. Prepared for U.S. Army Operational Evaluation Command, 1995.

Submitted/In Preparation

A. Hollis, R. Smith, A. Wilson, K. Pazdernik. Efficient, Well-Calibrated, and Interpretable Uncertainty Quantification for Deep Learning Classification. Revised and resubmitted to *Neural Processing Letters*.

A. Hollis, A. Wilson, R. Smith, G. Tompkins. Bayesian Hidden Markov Models for Arms Control Monitoring. Revised and resubmitted to *Military Operations Research Journal*.

AWARDS

Awards for National Security Impact

American Statistical Association Section on Statistics in Defense and National Security Distinguished Achievement Award (2018), *for outstanding accomplishment and sustained contributions at the intersection of the statistical profession and national defense*

Army Wilks Memorial Award (2015), *for contribution to the advancement of scientific or technical knowledge in Army statistics, ingenious application of such knowledge, or successful activity in the fostering of cooperation in scientific matters that benefit the Army, the Department of Defense, and the nation*

American Statistical Association Statistical Partnerships Among Academe, Industry, and Government (SPAIG) Award (2017), *recognizing outstanding partnerships between academe, industry, and government organizations, for the Laboratory for Analytic Sciences*

Department of Energy Defense Programs Award of Excellence (2009), *for significant contributions to the Stockpile Stewardship Program through contributions to the Joint Munitions Program*

Department of Energy Defense Programs Award of Excellence (2007), *for significant contributions to the Stockpile Stewardship Program through contributions to understanding high explosive detonator powder longevity*

Department of Energy Defense Programs Award of Excellence (2006), *as member of the Los Alamos National Laboratory Core Surveillance Transformation Team*

Department of Energy Defense Programs Award of Excellence (2006), *for significant contributions to the Stockpile Stewardship Program through initial surveillance transformation and strategy development for eliminating the legacy testing backlog*

Los Alamos National Laboratory Director's Distinguished Performance Award, Large Team (2007), *Detonator Powder Aging*

Los Alamos National Laboratory Achievement Award (2005), *for leadership in the quality initiative to transform reliability assessment for the nuclear stockpile*

Los Alamos National Laboratory Achievement Award (2000), *for leadership in DoD program development*

Honors and Recognition

Elected Member, Virginia Academy of Science, Engineering, and Medicine (2026)

Fellow of the American Association for the Advancement of Science (2015)

Elected Member of the International Statistical Institute (2012)

Fellow of the American Statistical Association (2008)

North Carolina State University Research Leadership Academy (2017), *for the university's most outstanding researchers and mentors*

2016-2017 Alumni Association Outstanding Research Award (2017)

Savage Dissertation Award in Bayesian Statistics (1995), *for Statistical Models for Shapes and Deformations*

North Carolina American Statistical Association Senior Statistician Award (2019), *for outstanding contributions to the theory and practice of statistics, particularly for contributions to the development of the subject in North Carolina*

NCSU Department of Statistics D.D. Mason Award (2019), *for Faculty Excellence*

Division of Academic and Student Affairs Thank an Advisor (2023), *recognizing outstanding advisors who go the extra mile to shape students through their mentorship and dedication*

Office of Faculty Excellence Thank a Teacher (2021), *recognizing outstanding faculty going the extra mile to shape students through their teaching and mentorship*

Office of Faculty Excellence Thank a Teacher (2013), *recognizing outstanding faculty going the extra mile to shape students through their teaching and mentorship*

LANL Star Award, Los Alamos Women's Diversity Working Group and the Office of Equal Opportunity and Diversity (2008)

Best Broad-Appeal Paper (with S. Keller-McNulty) for *Uncertainty Quantification for Complex Engineered Systems*, JANNAF 39th Combustion/27th Airbreathing Propulsion/21st Propulsion Systems Hazards/3rd Modeling and Simulation Joint Subcommittee Meeting, Colorado Springs, CO (2003)

Honorable Mention, Gertrude Cox Scholarship, American Statistical Association's Committee on Women in Statistics and the Caucus for Women in Statistics (1993)

Clare Boothe Luce Fellowship for Women in Science, Duke University (1992)
James B. Duke Fellowship in Statistics, Duke University (1992)
National Institute of Health, Division of Computer Research and Technology Employee of the Month (1992)
National Science Foundation Graduate Research Fellowship (1989)
Phi Beta Kappa (1989)

EXTERNAL FUNDING

Principal or co-principal investigator on over \$140M in sponsored funding

Research Funding

2013-24 Laboratory for Analytic Sciences, National Security Agency and other U.S. and International Intelligence Community Partners, \$94M (PI)
2023-24 Planning Grant for using Generative Artificial Intelligence in the Intelligence Community, \$100K (overall), \$49K (Wilson, Co-Investigator)
2021-22 Ammunition Resource Optimizer, Army Research Laboratory, \$234,000 (PI)
2018-22 Science of Test Research Consortium, Director Operational Test and Evaluation and Developmental Test and Evaluation Test Resource Management Center, \$478,430 (PI)
2020-21 Reliability Modeling and Experimental Design for the Extended Range Cannon Artillery Increment 2, U.S. Military Academy at West Point, \$20,000 (PI)
2016-21 NRT-DESE: Interdisciplinary Research Traineeships in Data-Enabled Science and Engineering of Atomic Structure, \$2.99M (Co-Investigator)
2019-20 MAVEN Test and Evaluation Research, Carnegie Mellon University, \$250,000 (PI)
2014-20 Consortium for Nonproliferation Enabling Capabilities, National Nuclear Security Administration, \$24M (overall), \$300,000/year (Wilson, Co-Investigator)
2017-19 Dose-Response Modeling of Existing Sonic Boom Datasets, National Institute of Aerospace for NASA, \$95,000 (PI)
2014-18 The Science of Test: Advanced Test and Evaluation in Support of the DOD Test and Evaluation Enterprise, Director Operational Test and Evaluation and Developmental Test and Evaluation Test Resource Management Center, \$6.3M (overall), \$100,000/year (Wilson, Co-Investigator)
2014-17 North Carolina Data Sciences and Analytics Initiative, UNC-General Administration, \$2.17M (overall), \$721,000 (NCSU, Co-PI)
2014-16 PARAKEET: Performance and Reliability Assessments for Knowledge of Electronic Emissions Technology, National Nuclear Security Administration, \$10.5M (overall), \$40,000/year (Wilson, Co-Investigator)
2013-15 Science of Security Lablet, National Security Agency, \$2.5M (overall), \$50,000/year (Wilson, Co-Investigator)
2014 Data Readiness Level, Laboratory for Analytic Sciences, \$100,000 (PI). Technical Coordinator for Data Readiness Task.
2013-14 Information Content to Assess Data Readiness, Laboratory for Analytic Sciences, \$50,000 (overall), \$25,000 (Wilson, Co-PI)
2013-14 Scalable Clustering Methods for Dynamic Health Data, Research Innovation Seed Funding, North Carolina State University, \$26,000 (Co-PI)

- 2013 Data Collection Instrument to Evaluate the Scientific Impact of Logistical Support Provided by the United States Antarctic Program, National Science Foundation, \$150,000 (IDA project, PI)
- 2013 Bibliometrics and Expert Judgment Methodology, IDA Corporate Research Program, \$15,000 (PI)
- 2013 Early Detection of Reliability Growth Issues, IDA Corporate Research Program, \$15,000 (PI)
- 2012-13 Dynamic Analysis and Management of CBRNE Collection Assets, Defense Threat Reduction Agency, \$400,000 (IDA project, PI)
- 2010-13 Statistically Significant Relational Data Mining, Sandia National Laboratories, \$1,699,000 (overall), \$265,000 (Wilson, Co-Investigator)
- 2012 Formalizing Observational Studies and Expert Panels, IDA Corporate Research Program, \$60,000 (Co-PI)
- 2008-12 Estimative Language and Statistical Relational Learning, Sandia National Laboratories, \$285,000 (Co-PI)
- 2005-08 LANL Systems Main Technical Element, Department of Energy Enhanced Surveillance Campaign, \$2.5M/year (LANL project, PI)
- 2003-08 Munitions Stockpile Reliability Assessment, Department of Defense/Department of Energy Joint Munitions Program, \$600,000/year (LANL project, PI)
- 2004 Ballistic Missile Defense System On-Alert Capability Assessment, Missile Defense Agency, \$150,000 (LANL project, PI)
- 2002-04 Space and Missile Defense Command Critical Measurements Program, Missile Defense Agency, \$300,000/year (LANL project, PI)
- 2001-02 Stores Separation Project, F-22 Program Office, \$300,000/year (LANL project, PI)
- 2001 Software Reliability Assessment, Army Test and Evaluation Command, \$100,000 (LANL project, PI)

Teaching Funding

- 2016 A Department-Wide Mentoring Initiative to Encourage URM Student Success in Statistics, North Carolina State University Diversity Mini-Grant Program, \$1600 (Co-Investigator)
- 2014 Workshop on Opportunities for Modern Statistics in Materials and Chemistry Research, Eastman Chemical Company, \$9000 (Co-PI)
- 2014 North Carolina State University Library Alt-Textbook Mini-Grant, \$500 (PI)
- 2011 Mathematics and Statistics Opportunities for Undergraduates Conference, Iowa State University Women's Enrichment Mini-Grant, \$800 (Co-PI)
- 2010-11 Strengthening Quantitative Literacy Using Case-Based Learning, Iowa State University Miller Faculty Fellowship, \$9672 (Co-PI)

TEACHING

University

- F2015 **ST 495/CSC 495 Introduction to Data Sciences** (46 students, 3 Cr.), North Carolina State University undergraduate course. Co-taught with R. Chirkova. Guest lecturer for Statistics Bootcamp in 2018, 2020, 2022, 2024, 2025.

- F2014 **ST 495/CSC 495 Introduction to Data Sciences** (49 students, 3 Cr.), North Carolina State University undergraduate course. Co-taught with R. Chirkova.
- F2013 **ST 740 Bayesian Inference and Analysis** (34 students, 3 Cr.), North Carolina State University graduate course.
- F2013 **ST 515 Experimental Statistics for Engineers I** (56 students, 3 Cr.), North Carolina State University graduate course. *Thank-a-Teacher Recipient*.
- F2012 **STAT 6289 Bayesian Computation** (10 students, 3 Cr.), George Washington University graduate course.
- S2011 **STAT 544 Bayesian Statistics** (19 students, 3 Cr.), Iowa State University graduate course.
- S2010 **STAT 544 Bayesian Statistics** (32 students, 2 sections, 3 Cr.), Iowa State University graduate course.
- F2009 **STAT 326 Introduction to Business Statistics II** (146 students, 4 sections, 3 Cr.), Iowa State University undergraduate course. Co-coordinated with U. Genschel (258 students, 7 sections).
- S2009 **STAT 544 Bayesian Statistics** (28 students, 3 Cr.), Iowa State University graduate course.
- S2009 **LAS 290G Special Problems/Understanding Technical Risk: Culture and Analysis** (5 students, 1 Cr.), Iowa State University undergraduate course. Co-taught with G. Wilson.
- F2008 **STAT 101 Principles of Statistics** (87 students, 2 sections, 4 Cr.), Iowa State University undergraduate course.
- S1995 **STA 290 Case Studies in Applied Statistics**, Duke University graduate course. Co-taught with M. Lavine.

Short Courses

- 2008 A. Wilson, C. Anderson-Cook, M. Hamada, S. Vander Wiel, E. Lawrence. Statistical Methods for Core Surveillance, Los Alamos National Laboratory, Los Alamos, NM
- 2004 C. Anderson-Cook, A. Wilson, A. Koehler, T. Graves, C. Chiu. Integrated Reliability Assessment. U.S. Army Aviation and Missile Command, Huntsville, AL
- 2004 A. Wilson. Reliability Section Joint Nuclear Explosives Training Facility Surveillance Engineering Course, Los Alamos National Laboratory, Los Alamos, NM
- 2003 S. Keller-McNulty, A. Wilson, G. Wilson, A. Koehler. Information Integration Technologies for Complex Systems. Institute for Mathematics and Its Applications, University of Minnesota, Minneapolis, MN
- 2003 S. Keller-McNulty, N. Hengartner, A. Wilson, C. S. Reese. Integrated Reliability Assessment. Marine Corps Program Department, Fallbrook, CA
- 2002 A. Wilson, D. Leishman, C. S. Reese. Knowledge Integration for Decision Making. Army SMART Conference, Salt Lake City, UT
- 2000 A. Wilson, C. S. Reese, S. Keller-McNulty, M. Meyer, J. Booker, T. Bement, M. McNulty. Data, Knowledge, and Information Integration to Support Decision Making. U.S. Army Conference on Applied Statistics, Rice University, Houston, TX
- 1992 A. Wilson. DCRT 213: SAS Fundamentals II. National Institutes of Health, Bethesda, MD
- 1992 A. Wilson. DCRT 212: SAS Fundamentals I. National Institutes of Health, Bethesda, MD

- 1992 A. Wilson, F. Yamada. DCRT 256: Topics in S-PLUS. National Institutes of Health, Bethesda, MD

Educational Outreach

- 2003- National Youth Science Camp, Bartow, WV
2025 Judge, North Carolina International Science Challenge
2018-23 Professional Strategies Working Group, Department of Statistics, NC State University, Raleigh, NC
2020 Teamwork, Developing Business and Professional Skills for Graduate Students and Postdocs Virtual Event, Duke University, Durham, NC
2020 Harnessing Data for Decision Making, Office of Undergraduate Research Connect Power Hour, NC State University, Raleigh, NC
2020 Data Science, Discussion for Faculty and Students from Nagoya University, NC State University, Raleigh, NC
2018 Statistics and Big Science, Keynote, NC Science Olympiad, Raleigh, NC
2018 Women in Data Science Panel Discussion, WiDS Raleigh @ NC State, Raleigh, NC
2014 Interdisciplinary Collaborations between Government, Industry, and Academia. Korean-American Scientist and Engineering Association, North Carolina Chapter.
2011 EDGE@ISU presents Mathematics and Statistics Opportunities for Undergraduates. Iowa State University, Ames, IA
2010 Preparing Future Faculty: Returning to Academia. Iowa State University, Ames, IA.
2010 Statistics in Defense and Security. Iowa Alliance Mathematics/ Statistics Research Experience for Undergraduates, Iowa State University, Ames, IA
2008 National Alliance for Doctoral Studies in the Mathematical Sciences, Second Annual Iowa Mathematical Field of Dreams Conference. Iowa State University, Ames, IA
2006 Statistics and Big Science. StatFest 2006, University of Texas, El Paso

ADVISING AND MENTORING

Faculty

- 2016-25 NCSU Cluster Faculty External Mentor
2014-21 NCSU Department Faculty Research and Teaching Mentor
2016 NCSU Mentor-Ring Mentor

ORI Faculty Fellows

- 2022-23 Daniela Jones
2020-22 Emily Griffith

Postdoctoral Fellows

- 2019-20 Christine Brugh
2017-19 Eli Typhina
2016-18 Héctor Rendón
2015-17 Karl Pazdernik

Ph.D. Students

- 2023 Conor Artman, North Carolina State University, Department of Statistics, “Artificial Intelligence for Medicine and Social Good”
- 2023 Andrew Hollis, North Carolina State University, Department of Statistics. “Statistical Methods for Nonproliferation and Counterproliferation” (co-advisor R. Smith)
- 2022 Laura Wendelberger, North Carolina State University, Department of Statistics, “Robust Statistical Methods for Model Selection and Land Cover Change Monitoring” (co-advisor B. Reich)
- 2018 Milo Page, North Carolina State University, Department of Statistics, “Automated Data Imputation: Extending Low-Rank Matrix Imputation Techniques For Statistical Prediction Modeling” (co-mentored by C. Gotwalt)
- 2017 Jordan Bakerman, North Carolina State University, Department of Statistics, “Twitter Analytics: Geotag Imputation, Forecasting, and Dynamic Variable Selection” (co-mentored by K. Pazdernik)
- 2017 Yiqing Tian, North Carolina State University, Department of Statistics, “Variable Selection in Logistic Regression with Applications” (co-advisor H. Bondell)
- 2016 Xiang Zhang, North Carolina State University, Department of Statistics, “Contributions to Statistical Methods for High-Dimensional and Dependent Data” (co-advisor L. Li)
- 2016 Zhen Han, North Carolina State University, Department of Statistics, “Statistical Methods for Relational Data: Visualization, Classification, and Topic Modeling on Networks”
- 2012 Roger Zoh, Iowa State University, Department of Statistics, “Using the Negative Log-Gamma Distribution for Bayesian System Reliability Assessment”
- 2011 Jiqiang Guo, Iowa State University, Department of Statistics, “Bayesian Methods for System Reliability and Community Detection”
- 2008 Kari Sentz, Binghamton University, Department of Systems Science and Industrial Engineering, “Methods of Probability and Imprecise Probability for Uncertainty Quantification in Applied Problems” (co-major professor G. Klir)

M.S. Students

- 2019 Jasme Lee, North Carolina State University, Department of Statistics, “Dose-Response Modeling of Quiet Sonic Boom Community Response Survey Data” (co-mentored by J. Rathsam)
- 2015 Hillary Graham, North Carolina State University, Department of Statistics, “Incorporating Concomitant Medications into Linear Models for Genome-Wide Association Studies” (co-mentored by D. Rotroff)
- 2011 Susan Vander Plas, Iowa State University, Department of Statistics, “Statistical Analysis of Atom Probe Mass Spectra”
- 2011 Matt Simpson, Iowa State University, Department of Statistics, “A Bayesian Analysis of an Exchange and Specialization Experiment”
- 2011 Erin Buchanan, Iowa State University, Department of Statistics, “Credible Intervals: Practical Applications in Energy Efficiency Impact Evaluation”
- 2011 Jing Li, Iowa State University, Department of Statistics, “Test Plans for Weibull Bayesian Assurance Testing in Reliability”
- 2010 Ralph Culver, Iowa State University, Department of Statistics, “Bayesian Analysis of Gauge R&R Data”

- 2010 Robert Foster, Iowa State University, Department of Statistics, “Simulation Analysis of a Bayesian Test Plan for Sequential Data from a Homogeneous Poisson Process”
- 2010 Randy Griffiths, Iowa State University, Department of Statistics, “Using Hierarchical Models to Find Test Plans for Binary Data”
- 2009 Roger Zoh, Iowa State University, Department of Statistics, “Comparison of Various Methods for Constructing Confidence Intervals for Two Multivariate Capability Indices in a Gauge R&R Study”

Ph.D. Committee Member

- 2025 Anil Radhakrishnan, North Carolina State University, Department of Physics
- 2025 Hyoshin Kim, North Carolina State University, Department of Statistics
- 2024 Khuzaima Hameed, North Carolina State University, Department of Statistics
- 2023 Kasia Dobryzcka, North Carolina State University, Department of Statistics
- 2021 Bowen Liu, North Carolina State University, Department of Statistics
- 2021 Matthew Miller, North Carolina State University, Department of Statistics
- 2020 Katherine Allen Moyer, North Carolina State University, Department of Statistics
- 2020 Joyce Cahoon, North Carolina State University, Department of Statistics
- 2019 Arnab Chakraborty, North Carolina State University, Department of Statistics
- 2019 Bharat Balagopal, North Carolina State University, Department of Electrical and Computer Engineering
- 2019 Isaac Michaud, North Carolina State University, Department of Statistics
- 2018 Adam Amos-Binks, North Carolina State University, Department of Computer Science
- 2018 Jami Jackson Mulgrave, North Carolina State University, Department of Statistics
- 2018 Siddharth Rastogi, North Carolina State University, Textile Technology Management
- 2018 Joshua Day, North Carolina State University, Department of Statistics
- 2017 Brian Naughton, North Carolina State University, Department of Statistics
- 2016 Kelsey Gaisor, North Carolina State University, Biomathematics Graduate Program
- 2016 Priyam Das, North Carolina State University, Department of Statistics
- 2016 Yuan Zhang, North Carolina State University, Department of Electrical and Computer Engineering
- 2016 Sam Morris, North Carolina State University, Department of Statistics
- 2016 Reka Howard, Iowa State University, Department of Statistics
- 2016 Yan Dora Zhang, North Carolina State University, Department of Statistics
- 2015 Ryan Parker, North Carolina State University, Department of Statistics
- 2015 Bradley Turnbull, North Carolina State University, Department of Statistics
- 2015 Prithwish Bhaumik, North Carolina State University, Department of Statistics
- 2014 Emily Casleton, Iowa State University, Department of Statistics
- 2013 Jia Liu, Iowa State University, Department of Statistics
- 2013 Shiyao Liu, Iowa State University, Department of Statistics
- 2013 Amy Hoeksema, Iowa State University, Department of Statistics
- 2012 Karl Pazdernik, Iowa State University, Department of Statistics
- 2011 Cherie Alf Kientoff, Iowa State University, Department of Statistics
- 2011 Brian Weaver, Iowa State University, Department of Statistics
- 2010 Adam Pintar, Iowa State University, Department of Statistics

M. S. Committee Member

- 2015 Kaaviya Palanivel Kathirvel, North Carolina State University, Integrated Manufacturing and Systems Engineering Institute
- 2014 Andrea Kaplan, Iowa State University, Department of Statistics
- 2010 Luke Fostvedt, Iowa State University, Department of Statistics
- 2010 Shiyao Liu, Iowa State University, Department of Statistics
- 2010 Amy Borgen, Iowa State University, Department of Statistics
- 2008 Brian Weaver, Iowa State University, Department of Statistics

Graduate Research Assistants

- 2021-22 Kate Sanborn, Hyoshin Kim
- 2020-21 Andrew Hollis, Kate Sanborn, Khuzaima Hameed
- 2019-20 Andrew Hollis, Joyce Cahoon, Chris Edwards
- 2018-19 Katherine Allen Moyer, Andrew Hollis, Jasme Lee, Todd Wilson, Caleb Weaver, Joyce Cahoon
- 2017-18 Katherine Allen Moyer, Jordan Bakerman, James Gilman, Nick Johnson, Jasme Lee, Susheela Singh, Laura Wendelberger
- 2016-17 Jordan Bakerman, James Gilman, Susheela Singh
- 2015-16 Jordan Bakerman, Caleb Browning, James Gilman, Hillary Graham, Zhen Han, Nick Meyer, Xiang Zhang
- 2014-15 Caleb Browning, Nick Meyer, Merve Tekbudak
- 2013-14 Andrea Kostura

Undergraduate Research Assistants

- 2019-20 Kate Sanborn
- 2014-15 James Gilman

NC State Park Scholar Mentor

- 2021-23 Anika Bhadriraju
- 2019-23 Katelyn McInerney
- 2018-22 Shannon Pinnell

NC State Consulting Project Mentor

- 2016 Candice Park and Tian Guo, Survival Analysis of Faculty Retention
- 2014 Jennifer Ran Wei and Qianwen Tan, NC 4-H Evaluation Learning Circle

LANL Summer Students

- 2007 Brian Weaver, Iowa State University
- 2006 Brian Weaver, University of New Mexico
- 2003 Kristi O'Grady, North Carolina State University
- 2003 Morgan Harris, Virginia Military Institute
- 2001 Meng Xu, University of Idaho
- 2001 Andrew Swift, George Washington University

SERVICE

National Academies of Sciences, Engineering, and Medicine

- 2023-24 Committee on Statistical and Data-Driven Methods for Additive Manufacturing Qualification--Future Directions and Opportunities in Materials and Models: A Workshop
- 2021-24 Committee for Emerging Trends in Aviation Safety
- 2018-24 Committee on Applied and Theoretical Statistics
- 2021-23 Committee on Risk Analysis Methods for Nuclear War and Nuclear Terror
- 2021 Panel on Review of Selected Divisions of the National Institute of Standards and Technology Information Technology Laboratory
- 2019-21 Committee for Improving Defense Acquisition Workforce Capability in Data Use
- 2017 Committee on Learning from the Science of Cognition and Perception for Decision Making: A Workshop
- 2012-14 Committee on Integrating Humans, Machines and Networks: A Global Review of Data-to-Decision Technologies
- 2011-12 Committee to Review the Testing of Body Armor Materials for Use by the U.S. Army, Phase 3
- 2010-12 Committee on Mathematical Foundations of Validation, Verification, and Uncertainty Quantification
- 2008-11 Panel on Industrial Methods for the Effective Testing and Development of Defense Systems
- 2010 Committee to Review the Testing of Body Armor Materials for Use by the U.S. Army, Phase 2
- 2009 Committee to Review the Testing of Body Armor Materials for Use by the U.S. Army, Phase 1
- 2006-07 Panel on Methodological Improvement to the Department of Homeland Security's Biological Agent Risk Analysis
- 2002-03 Panel on the Operational Test Design and Evaluation of the Interim Armored Vehicle

National Laboratory Review Appointments

- 2024-27 University of Tennessee Oak Ridge Innovation Institute External Review Board
- 2026 Lawrence Livermore National Laboratory, Decision Superiority Program Review Committee
- 2022-26 Los Alamos National Laboratory Science of Signatures (SoS) Capability Review Committee, Chair (2025-2026)
- 2023-25 Pacific Northwest National Laboratory National Security Directorate Advisory Committee
- 2024 Pacific Northwest National Laboratory, Framework for Evaluating Radical New (Safeguards) Scenarios Review Panel
- 2021-22 Pacific Northwest National Laboratory Chemical Dynamics Initiative Review Committee
- 2008-18 Sandia National Laboratories, Predictive Engineering Science Panel

Advisory and Review Appointments

- 2026 Air Force Institute of Technology, Data Science and Operations Research Programs Review Committee

2024-25 North Carolina Board of Science, Technology, and Innovation
 2023-25 U.S. Military Academy at West Point Applied Statistics and Data Science Advisory Board
 2023 University of Virginia, Biocomplexity Institute Review Committee
 2022 Department of Energy, Office of Defense Nuclear Nonproliferation Independent Review for Next-Generation AI for Next-Generation Nuclear Assessment
 2022 Department of Energy, Office of Defense Nuclear Nonproliferation Independent Review of Sequential Bayesian Approach to Warhead Verification
 2017-21 National Center for Atmospheric Research, Computational Information Systems Laboratory Advisory Panel
 2020 Brigham Young University, External Review Committee, Department of Statistics
 2013-15 Bowling Green State University, Department of Mathematics and Statistics Computational Mathematics and Statistics Advisory Board
 2013 Department of Energy, Applied Mathematics Program Committee of Visitors
 2013 Department of Energy, Office of Science Advanced Scientific Computing Research Crosscutting Requirements Review
 2009-11 Department of Energy, University of Michigan Predictive Science Academic Alliance Program Review Panel
 2010 Invited Participant, Defense Threat Reduction Agency/Sandia National Laboratories Workshop on Challenges in Computational Social Science
 2008 Department of Energy, Office of Science Applied Mathematics Research Program, Workshop on Mathematical Issues for Petascale Data Sets, Organizing Committee and Invited Participant
 2008 Naval Postgraduate School, Chief of Naval Operations Distinguished Fellows Workshop on Critical Infrastructure Vulnerability, Naval Postgraduate School
 2006 Department of Energy, Office of Science Workshop on Mathematical Research Challenges in Optimization of Complex Systems
 2006 Department of Energy, Office of Nuclear Energy and Office of Advanced Scientific Computing Research, Simulation and Modeling for Advanced Nuclear Energy Systems Workshop, Sub-committee Co-Chair

Institutional Service

William & Mary

2026- Board, Virginia Space Grant Consortium
 2026- Councilor, Oak Ridge Associated Universities
 2026- Representative, Southeastern Universities Research Association
 2026 Executive Board, Omohundro Institute
 2026 University Professor for Research Excellence Award Committee

North Carolina State University

2025 Strategic Information Technology Committee
 2024-25 University of Tennessee-Battelle Oak Ridge National Laboratory Board of Governors
 2024-25 Renaissance Computing Institute (RENCI) Oversight Board
 2024-25 Triangle Universities Center for Advanced Studies Inc. (TUCASI) Board of Trustees
 2024 Search Committee, Assistant Vice Chancellor for Research Computing and Data
 2024 Coordinator, Research Administration and Support Services Task Forces

2023-25 Research, Scholarship, and Creativity Information Technology Committee
 2023-25 Chair, Steering Committee, Advanced Research and Test Reactor Feasibility Study
 2020- Research Data Security Compliance Steering Group
 2023 Research Computing Improvement Task Force
 2023 Data Storage Strategic Task Force
 2023 Organizing Committee, University Research Symposium on Putting Data Science into Practice: Emerging Role of AI
 2022-23 Search Committee, Executive Director for the Friday Institute of Educational Innovation and Associate Dean for Translational Research, College of Education
 2022-23 Search Committee, Faculty Position in Artificial Intelligence, College of Agriculture and Life Sciences
 2022-23 Search Committee, Director of the Office of Undergraduate Research
 2021-22 Analytics Hub Steering Team
 2019-22 University Standing Committee on Commencement
 2021 Research Task Force Co-Chair, Post COVID-19 Innovation Task Force
 2021 Search Committee Chair, Data Science Academy Executive Director
 2021 Search Committee Chair, Data Science Academy Teaching Postdoctoral Fellows
 2017-21 GLBT Advocate Program
 2020 Research Computing and Data Service Team
 2019-20 Search Committee Chair, Data-Driven Science Cluster
 2014-19 Director (2019), Associate Director (2014-2018), Data Science Initiative, Office of Research and Innovation
 2015-17 Associated Faculty, Department of Computer Science
 2016 Task Force for Interdisciplinary and Joint Faculty Review
 2014-25 Cluster Coordinator, Chancellor's Faculty Excellence Program in Data-Driven Science
 2014-17 University Standing Committee on Commencement
 2014-17 University Diversity Advisory Committee
 2013-16 Search Committee, Data-Driven Science Cluster

North Carolina State University, Department of Statistics

2019 Qualifying Exam Committee
 2018-19 Faculty Search Committee
 2017-18 Seminar Committee
 2016-17 Department Head Search Committee
 2014-15 Big Data Committee
 2014-15 Faculty Search Committee
 2013-15 Preliminary Exam Committee (Chair, 2014-2015)

Institute for Defense Analyses

2013-19 Institute for Defense Analyses Adjunct Research Staff Member and Faculty Fellow
 2012 Baccalaureate Fellows Hiring Committee
 2012 Research Associate Hiring Committee
 2012 Research Staff Hiring Committee

Iowa State University, College of Liberal Arts and Sciences

2009-11 College Diversity Committee

2010-11 Senior Mentor, EDGE@ISU Mentoring Cluster (Enhancing Diversity in Graduate Education, Departments of Mathematics and Statistics)

Iowa State University, Department of Statistics

2011-13 Collaborating Associate Professor, Department of Statistics, Iowa State University

2010-11 Graduate Committee

2011 Seminar Chair

2010-11 Department Diversity Representative

2009-10 Curriculum Committee

2009-10 PhD/MS Exam Committee

2008-11 Iowa STAT-ers Advisor

2008-10 Faculty Search Committee

2008-09 Information Technology/System Administrator Search Committee

2008-09 Computation Advisory Committee

Los Alamos National Laboratory

2008- Guest Scientist, Statistical Sciences Group, Los Alamos National Laboratory

2000-08 Technical Lead for DoD Programs, Statistical Sciences Group

2000-08 Director of Student Programs, Statistical Sciences Group

2007 Los Alamos National Laboratory Laboratory-Directed Research and Development
Exploratory Research in Computing and Information Sciences Peer-Review and
Funding Committee

2005 Statistical Sciences Group Leader Search Committee

2003 Los Alamos National Laboratory Fellows Selection Committee

2003 Nuclear Design and Risk Analysis Group Leader Search Committee

2000-02 Los Alamos National Laboratory Laboratory-Directed Research and Development
Exploratory Research in Mathematics, Simulation, and Modeling Peer-Review and
Funding Committee

Professional Societies

2027-29 American Statistical Association, Incoming Chair/Chair/Past Chair, Council of Sections

2024-26 Canadian Statistical Sciences Institute Scientific Advisory Committee

2024-26 American Statistical Association, Samuel S. Wilks Memorial Awards Committee

2018-25 National Institute for Statistical Sciences Board of Trustees, Chair (2023-2025),
Executive Committee (2021-2025), Finance Committee (2018-2020, 2023-2025)

2021-25 International Society for Business and Industrial Statistics Council, International
Statistical Institute

2023 Society for Industrial and Applied Mathematics Task Force on the Future of
Computational Science

2019-21 American Statistical Association, Vice Chair, Council of Sections

2016-20 American Association for the Advancement of Science, Secretary, Section U
(Statistics)

2013-18 American Statistical Association, Council of Sections Representative, Section on
Statistics in Defense and National Security

2010-12 American Statistical Association, Nomination Committee, Section on Bayesian
Statistical Sciences

- 2009-11 American Statistical Association, *Techometrics* Management Committee
- 2009-11 American Statistical Association, Program Chair-Elect, Program Chair, Past Program Chair, Section on Bayesian Statistical Sciences (2010 JSM Program Committee)
- 2009-10 American Statistical Association Working Group on Strategic Plan Focus: Visibility and Impact in Policy Making
- 2008-10 American Statistical Association, Speakers Committee Chair, Section on Statistics in Defense and National Security
- 2006-08 American Statistical Association, Chair-Elect, Chair, Past Chair, Section on Statistics in Defense and National Security
- 2006 Chair, American Statistical Association President's Task Force on Statistics in Defense and National Security
- 2004-05 American Statistical Association, Secretary/Treasurer/Webmaster, Section on Statistics in Defense and National Security
- 2002-03 American Statistical Association, President, Albuquerque Chapter
- 2001-02 American Statistical Association, Vice-President, Albuquerque Chapter
- 2000-05 American Statistical Association, Committee on Statisticians in Defense and National Security

Conference

- 2024 Organizer and Technical Program Committee, Oak Ridge National Laboratory Core Universities Artificial Intelligence Workshop
- 2022-23 Singpurwalla Memorial Symposium on New Frontiers in Reliability and Risk Analysis Organizing Committee
- 2018-23 Defense and Aerospace Test and Analysis Workshop (DATAWorks), Technical Program Committee (2020-2023), Co-Chair Committee (2019), ASA/SDNS General Chair (2018)
- 2019 Organizer, Data Science and Big Data Workshop, University and Global Partnership Network, Raleigh, NC
- 2017 Advisory Committee, Science of Test Workshop
- 2000-16 Army Conference Applied Statistics/Conference on Applied Statistics in Defense, Executive Committee (2000-2016), Organizing Committee Chair (2016), Program Chair (2014), Local Organizer (2001)
- 2016 Organizer, Invited Panel, Public-Private Partnerships to Accelerate Innovation in Intelligence Analysis, American Association for the Advancement of Science Meeting, Washington, DC
- 2015 Organizing Committee, Institute for Computational and Experimental Research in Mathematics (ICERM) Workshop on Data Science
- 2014 Organizing Committee, Department of Energy Conference on Data Analysis, Santa Fe, NM
- 2014 Organizing Committee, Women in Statistics Conference, Cary, NC
- 2012 Organizer, Invited Session, "Reliability for Complex Systems" for Quality and Productivity Research Conference, Long Beach, CA
- 2010 Program Committee, Joint Statistical Meetings, Vancouver, BC
- 2010 Program Committee, *Eastern North American Region*/International Biometric Society
- 2010 Organizer, Invited Session, "Bayesian Analysis Invited Session" for Joint Statistical Meetings, Vancouver, BC

- 2009 Program Committee, Sixth International Symposium on Imprecise Probabilities and Their Applications
- 2009 Organizer, Invited Session, “Quantitative Methods for Combating Bioterrorism” for Joint Statistical Meetings, Washington, DC
- 2007 Organizer, Invited Session, “Advances in System Reliability” for Joint Statistical Meetings, Salt Lake City, UT
- 2005 Organizer, Invited Session, “Experiment Planning Issues in System Reliability” for Design and Analysis of Experiments Conference, Santa Fe, NM
- 2005 Organizer, Invited Session, “Statistical Methods for Defense Applications” for Spring Research Conference on Statistics in Industry and Technology, Park City, UT
- 2004 Co-Organizer, Mathematical Methods in Reliability Conference, Santa Fe, NM
- 2002 Organizer, Invited Session, “Minimizing the Magic: Transforming Knowledge Representations into Statistical Models” for Spring Research Conference on Statistics in Industry and Technology, Ann Arbor, MI

Editorial

- 2018-21 Associate Editor, *Harvard Data Science Review*
- 2018 Guest editor for *Chance* special issue (April 2018) on defense and national security (with D. Banks)
- 2012-14 Associate Editor, *Journal of Uncertainty Quantification*
- 2011-13 Reviews Editor of the *Journal of the American Statistical Association* and *The American Statistician*
- 2010-12 Managing Editor, *Bayesian Analysis*
- 2008-10 Associate Editor for Reviews, *Journal of the American Statistical Association* and *American Statistician*
- 2006 Guest editor for *Statistical Science* special issue on reliability (with S. Keller-McNulty and C. Anderson-Cook)

Review and Referee

National Science Foundation, Department of Energy Office of Science, National Academies of Sciences, Engineering, and Medicine, Army Research Office, American Association for the Advancement of Science, John Wiley & Sons, Taylor & Francis, Springer, *Applied Stochastic Models in Business and Industry*, *Bayesian Analysis*, *Chinese Journal of Aeronautics*, *CRC Press*, *Harvard Data Science Review*, *IEEE Transactions on Engineering Management*, *IEEE Transactions on Fuzzy Systems*, *IEEE Transactions on Medical Imaging*, *International Journal of Approximate Reasoning*, *International Journal of General Systems*, *Journal of the American Statistical Association*, *Journal of Statistics Education*, *Mathematical Geology*, *Naval Research Logistics*, *Notices of the American Mathematical Society*, *Quality Engineering*, *Quality and Reliability Engineering International*, *Reliability Engineering and System Safety*, *Risk Analysis*, *South African Statistical Journal*, *Springer*, *Stat*, *Statistical Science*, *Statistics in Medicine*, *Technometrics*

UNREFEREED PUBLICATIONS

Book Reviews

- A. Wilson (2013). Web Development with SAS by Example, 3rd ed. *The American Statistician* 67(2): 113.
- A. Wilson (2010). Review of Bayesian Evaluation of Informative Hypotheses edited by H. Hoijsink, I. Klugkist, P. Boelen. *Journal of the American Statistical Association* 105(489): 438.
- A. Wilson (2009). Review of Bayesian Networks and Decision Graphs, 2nd Edition by F. Jensen and T. Nielsen. *Journal of the American Statistical Association* 104(485): 410.

Conference Proceedings

- N. Spruill, A. Wilson, E. Seglie (2008). Statistics in Defense and National Security: The Present—You Need More Than Statistics—You Need the Right Statistics. *Proceedings of the Section on Statistics in Defense and National Security*, American Statistical Association, Denver, CO
- A. Wilson, C. Anderson-Cook (2007). Analysis and Data Collection Strategies for Multilevel Complex Systems. *Proceedings of the 2007 Mathematical Methods in Reliability Conference*, Glasgow, UK
- N. Hengartner, A. Wilson (2004). Stockpile Reliability Assessment. *Proceedings of the 2004 Mathematical Methods in Reliability Conference*, Santa Fe, NM
- C. S. Reese V. Johnson, M. Hamada, A. Wilson (2004). A Hierarchical Model for System Reliability Based on Lifetime Data. *Proceedings of the 2004 Mathematical Methods in Reliability Conference*, Santa Fe, NM
- A. Wilson, S. Keller-McNulty (2003). Uncertainty Quantification for Complex Engineered Systems. *Proceedings of JANNAF 39th Combustion/27th Airbreathing Propulsion/21st Propulsion Systems Hazards/3rd Modeling and Simulation Joint Subcommittee Meeting*, Colorado Springs, CO
- A. Wilson (2002). Integrating Information Using Belief Functions. *Proceedings of Mathematical Methods in Reliability 2002*, Trondheim, Norway
- H. Martz, M. Hamada, C. S. Reese, A. Wilson (2002). Using Genetic Algorithms to Design Bayesian Reliability Experiments. *Proceedings of Mathematical Methods in Reliability 2002*, Trondheim, Norway
- J. Booker, S. Keller-McNulty, M. Meyer, C. S. Reese, A. Wilson (2001). Probabilistic Information Integration Technology. *Proceedings of the 6th Annual Conference on Turbine Engine High Cycle Fatigue*, Jacksonville, FL
- A. Wilson. Discussion of Practical Inferences for Decision Making in Industrial Applications (1997). *Proceedings of the Section on Statistical Education*, American Statistical Association, Anaheim, CA
- C. McCulloch, J. Laading, A. Wilson, V. Johnson (1996). A Shape-Based Framework for Automated Image Segmentation. *Proceedings of the Section on Bayesian Statistical Science*, American Statistical Association, Chicago, IL
- A. Wilson, V. Johnson, L. Yu, S. Pizer (1995). Scale-Space Image Models for Shape Deformation. *Proceedings of the 27th Symposium on the Interface: Computing Science and Statistics*, Interface Foundation of North America, Pittsburgh, PA

- A. Wilson, V. Johnson (1994). Using Features to Model Prior Structural Information. *Proceedings of the Section on Statistics in the Physical and Engineering Sciences*, American Statistical Association, Toronto, CA
- A. Wilson, J. Malley, J. Pfeifer, A. Petelin (1992). Remarks on the Gibbs Sampler and its Implementation on a Parallel Machine. *Proceedings of the Statistical Computing Section*, American Statistical Association, Boston, MA

Discussion Papers

- A. Wilson, M. Morris (1999). Statistical Methods to Validate the National Missile Defense Battle Management Command and Control Software: Summary of Presentation at U.S. Army Conference on Applied Statistics. LA-UR-99-5793.
- C. McCulloch, J. Laading, A. Wilson, V. Johnson (1996). A Shape-Based Framework for Automated Image Segmentation. Duke University ISDS Discussion Paper 96-31.
- A. Wilson, V. Johnson (1994). Priors on Scale-Space Templates. Duke University ISDS Discussion Paper 94-12.
- A. Wilson (1994). Cognitive Factors Affecting Subjective Probability Assessment. Duke University ISDS Discussion Paper 94-02.

PRESENTATIONS

Invited/Seminar

- 2026 Innovation in Workforce Development, Panelist, SURA 2026 Symposium on Innovation in Data Science Research, Jefferson Laboratory, Newport News, VA
- 2026 Shaping the Future of Conservation, Moderator for Panel with Derek Aday and Robert Rose, William & Mary Alumni Association, Richmond, VA
- 2026 The Next 250: Navigating the AI Revolution and Future of Democracy, Moderator for panel with Chiraag Bains, Tim Carroll, Sunita Patel, AI and Democracy Weekend, Williamsburg, VA
- 2025 Emerging Technology, Competition, and the Future of the Research University, Moderator for conversation with Dr. Melissa Flagg, Williamsburg, VA
- 2025 Small Data Problems in a Big Data World, Masters of Science Seminar, Idaho National Laboratory, Idaho Falls, ID
- 2025 Data-Driven Discovery, Plenary speaker at NC Department of Transportation Research and Innovation Seminar, Raleigh, NC
- 2024 Navigating Paths to Tenure and Promotion – and Pitfalls Along the Way, Invited Panel, Women in Statistics and Data Science Conference, Reston, VA
- 2024 From Theory to Practice: Transforming Analytics with Cutting-Edge Research, North Carolina State University Business Analytics Initiative Roundtable, Raleigh, NC
- 2024 Statistics, Data Analytics, and AI: A Primer, Workshop on Statistical and Data-Driven Methods for Additive Manufacturing Qualification, National Academy of Science, Engineering, and Medicine, Irvine, CA
- 2024 Artificial Intelligence at NC State, Defense Alliance of North Carolina Data Decoded: Defense Opportunities in North Carolina's AI Landscape, Raleigh, NC
- 2023 Science in the Age of Generative AI Panel, ORNL Core Universities AI Workshop, Georgia Tech, Atlanta, GA

- 2023 A Statistician's Journey: From Bayes to AI, ORNL Core Universities AI Workshop, Georgia Tech, Atlanta, GA
- 2023 AI@NC State Hot Topics: When do we stop? AI and Ethics, NC State Data Science Academy, Raleigh, NC
- 2023 *The Impact of Artificial Intelligence Technologies on Quality and Statistics*, Fall Technical Conference, Raleigh, NC
- 2023 Panel on Regional Leadership, Raleigh-Austin Intercity Visit, Raleigh, NC
- 2023 Lunchtime Panel on Collaboration, New Mexico American Statistical Association Annual Meeting, Santa Fe, NM
- 2023 Demystifying Data Science, New Mexico American Statistical Association Annual Meeting, Santa Fe, NM
- 2023 DoD Funding at NC State (two-webinar series), North Carolina State University
- 2023 Panel on Strategies for Planning, Placing, and Pitching Data Science Resources in Higher Education, webinar follow up to the Alfred P. Sloan Workshop on Modeling Data Science Consulting in Higher Education
- 2023 Information Environment for Research and NC State and the Laboratory for Analytic Sciences, Institute for Defense and Business
- 2023 Undergraduate Research, GRAD-Future Workshop, North Carolina State University
- 2023 Conversation about Foundational Ideas in Artificial Intelligence, University Research Symposium, North Carolina State University
- 2022 Research Leadership Academy Lunch and Learn on Exploring External Awards, North Carolina State University
- 2022 Data Science and Defense at North Carolina State University, STR Analytics Seminar Series (str.us)
- 2022 Data Science and Interdisciplinary Collaboration, Climate Informatics 2022, North Carolina Institute for Climate Studies, Asheville, NC
- 2022 Plenary Panel, Open Climate Data Science Workshop, North Carolina Institute for Climate Studies and North Carolina Climate Office
- 2022 Information Environment for Research and NC State and the Laboratory for Analytic Sciences, Institute for Defense and Business
- 2021 Plenary Welcome, NASA iTECH Cycle II Forum, North Carolina State University
- 2021 Facets of a Diverse Career, Plenary Speaker at JMP Discovery Summit Americas
- 2021 Plenary Panel on Diversifying Workforce, 3rd National Oceanic and Atmospheric Administration Artificial Intelligence Workshop
- 2021 Data Science at NC State, Data Economy Committee, Board of Science, Technology, and Innovation, Department of Commerce, State of North Carolina
- 2021 Data Science is for Everyone. 20th Annual Social Equity Leadership Conference, National Academy of Public Administration
- 2020 What is Big Data and What Does it Mean? Big Data and Artificial Intelligence in Toxicology, Genetics and Environmental Mutagenesis Society of North Carolina Fall Meeting
- 2020 Big Data: What is It? And What Does it Mean to Me? Webinar for the Scientific Liaison Coalition (<https://www.toxicology.org/slc.asp>), with Aric LaBarr
- 2020 Data Science Primer, Webinar for National Academies of Sciences, Engineering, and Medicine's Committee for Improving Defense Acquisition Workforce Capability in Data Use

- 2019 Demystifying Data Science, North Carolina Women in Machine Learning and Data Science, Research Triangle Park, NC
- 2019 Laboratory for Analytic Sciences Overview, Intelligence Community Academic Research Symposium, Washington, DC, with Jen Banwart
- 2019 Data Science and Analytics: What is It? Environmental Protection Agency, Raleigh, NC, with Aric LaBarr
- 2019 Combining Information to Assess the Reliability of Complex Systems, Joint Statistical Meetings, Denver, CO
- 2019 Data Science in Defense and National Security, Emerging Data Science Methods for Complex Biomedical and Cyber Data, Augusta, GA
- 2018 SPAIG Award Collaboration: Laboratory for Analytic Sciences, Joint Statistical Meetings, Vancouver, BC
- 2018 Partnering for Data Science: The Laboratory for Analytic Sciences, 2018 Symposium on Data Science and Statistics, Reston, VA
- 2018 Demystifying Data Science, Defense and Aerospace Test and Analysis Workshop (DATAWorks 2018), Springfield, VA
- 2017 Data Science at the Laboratory for Analytic Sciences, RTI Data Science Lunch and Learn, Research Triangle Park, NC
- 2017 Data Science at NC State, ORIED Thought Leaders Program on Data Science, Raleigh, NC
- 2017 Bayesian Methods for Defense and Security, NASA-Langley, Hampton, VA
- 2017 Data Analytics and National Security, National Nuclear Security Administration, Department of Energy, Washington, DC
- 2017 Panel on How to Introduce Management to Bayesian Reliability, Joint Statistical Meetings, Baltimore, MD
- 2017 Statistics and Data Science in Defense and Security, Smith College, Northampton, MA
- 2017 Introduction to Bayesian Statistics Tutorial, Science of Test Workshop, Springfield, VA
- 2017 Bayesian Analysis, Statistical Engineering Leadership Webinar
- 2016 Assurance Testing or Reliability, Quality and Productivity Research Conference, Phoenix, AZ
- 2016 Leading a Data Analytics Workforce, Panel Discussion, Cisco Data Science for Leaders, Video-Telepresence from Raleigh, NC
- 2016 Combining Information for Reliability Assessment, Rigorous Test and Evaluation for Defense, Aerospace, and National Security, Alexandria, VA
- 2016 Developing a Data-Driven Organization, Keynote Panel, Data 4 Decisions Conference and Exposition, Raleigh, NC
- 2016 Bayesian Reliability: Combining Information, Plenary Talk at Stu Hunter Research Conference, Waterloo, Ontario, CA
- 2016 Combining Information to Assess System Reliability, Sandia Statistical Sciences Colloquium Series, Albuquerque, NM
- 2016 Data Science Initiatives at North Carolina State University, Cisco Analytics Speaker Series, Video-Telepresence from Raleigh, NC
- 2015 Introduction to Data Science: An Interdisciplinary Course for Undergraduates, Joint Statistical Meetings, Seattle, WA
- 2015 Infusing Bayesian Thinking in Collaborative Projects, Roundtable Breakfast, Section on Physical and Engineering Sciences, Joint Statistical Meetings, Seattle, WA

- 2015 Combining Information for Uncertainty Quantification, INFORMS Roundtable, Jackson Hole, WY
- 2015 Estimating System Reliability from Heterogeneous Data, Mathematical Methods in Reliability, Tokyo, Japan
- 2015 Combining Information to Assess System Reliability, Department of Statistics, University of South Carolina, Columbia, SC
- 2014 Research at the Laboratory for Analytic Sciences, Virginia Bioinformatics Institute at Virginia Tech Arlington, VA
- 2014 Combining Information to Assess the Reliability of DoD Systems, Joint Research Conference on Statistics in Quality, Industry, and Technology, Seattle, WA
- 2014 Combining Information to Assess System Reliability, Data Mining in Business in Industry, International Symposium on Business and Industrial Statistics/Conference of the ASA Section on Statistical Learning and Data Mining, Durham, NC
- 2014 Panel on Working in Interdisciplinary Teams, Conference on Women in Statistics, Cary, NC
- 2014 Panel on Career Flexibility, Conference on Women in Statistics, Cary, NC
- 2014 Panel on Big Data: When More is Too Much, Chancellors Faculty Excellence Program Symposium, North Carolina State University, Raleigh, NC
- 2014 Combining Information in Engineering Statistics. College of Textiles, North Carolina State University, Raleigh, NC
- 2014 Bayesian Approaches to Combining Information for Reliability Assessment. IEEE Eastern North Carolina Section Seminar, Raleigh, NC
- 2013 Bayesian Methods in Defense and Security. Duke University, Durham, NC.
- 2013 Bayesian Reliability Applications in the Department of Defense. Quality and Productivity Research Conference, Niskayuna, NY
- 2013 Bayesian Models for Community Detection. Brown Bag Lunches at IDA Science and Technology Policy Institute and System Evaluation Division, Washington, DC and Alexandria, VA
- 2012 Bayesian Methods for Estimating System Reliability Using Heterogeneous Multilevel Information. Conference on Data Analysis, Santa Fe, NM (with Jiqiang Guo)
- 2011 Bayesian Reliability. Roundtable Luncheon, Section on Quality and Productivity, Joint Statistical Meetings, Miami, FL
- 2011 Collaborations Between Iowa State University and Los Alamos and Sandia National Laboratories. Joint Statistical Meetings, Miami, FL.
- 2011 Bayesian Methods for Complex Systems. Marketing Colloquium Distinguished Speaker Series, Iowa State University, Ames, IA
- 2010 Strengthening Quantitative Literacy Using Case-Based Learning. Problem-Solving Faculty Learning Community, Iowa State University, Ames, IA
- 2010 Statistical Challenges from Science-Based Stockpile Stewardship. Institute for Defense Analysis Center for Communications Research, La Jolla, CA
- 2010 Introduction to Bayesian Reliability. Fall Technical Conference, Birmingham, AL
- 2010 Bayesian Methods for Estimating the Reliability of Complex Systems using Heterogeneous Multilevel Data. University of Minnesota, Minneapolis, MN
- 2010 Assessing the Methodology for Testing Body Armor. Joint Statistical Meetings, Vancouver, BC

- 2010 Statistics Methods Used in Defense and Non-Defense Applications. Invited Panel, Joint Statistical Meetings, Vancouver, BC
- 2010 Bayesian Methods for Estimating the Reliability of Complex Systems using Heterogeneous Multilevel Data. Simon Fraser University, Vancouver, BC
- 2009 Planning Surveillance for a Stockpile that Might Degrade. Invited Poster, Joint Statistical Meetings, Washington, DC
- 2009 Terrorist Risk Assessment. Joint Statistical Meetings, Washington, DC
- 2009 Statistical Challenges from Science-Based Stockpile Stewardship. University of Iowa, Iowa City, IA
- 2009 Statistical Challenges from Science-Based Stockpile Stewardship. Texas A&M University, College Station, TX
- 2008 Bayesian Reliability Analysis. Interface 2008 Risk: Reality. Durham, NC
- 2008 Reliability Case Study. U. S. Military Academy Department of Mathematical Sciences
- 2008 Statistics Workshop: Real Data and the Art of Statistical Analysis, West Point, NY
- 2007 How DHS Currently Manages Risk. RISK: Policy, Perception, and Practice Workshop, Statistical and Mathematical Sciences Institute, Research Triangle Park, NC
- 2007 Joint Munitions Program Munitions Stockpile Reliability Assessment Project. Los Alamos National Laboratory Joint Munitions Program Seminar Series, Los Alamos, NM
- 2007 Increasing Understanding of the Need for Statistics in Defense and Security. Roundtable Luncheon, Section on Statistics in Defense and National Security, Joint Statistical Meetings, Salt Lake City, UT
- 2007 Risk Analysis at Los Alamos National Laboratory. Joint Statistical Meetings, Salt Lake City, UT
- 2006 Bayesian Reliability. Roundtable Luncheon, Joint Statistical Meetings, Seattle, WA
- 2006 Systems Reliability and Experiment Planning. Interface 2006, Pasadena, CA.
- 2006 Next Generation Stockpile Reliability Methods. National Nuclear Security Administration Technical Seminar, Washington, DC
- 2006 Next Generation Stockpile Reliability Assessment. National Nuclear Security Administration Surveillance Policy Integrated Requirements Council, Albuquerque, NM
- 2005 System Reliability and Experiment Planning. U.S. Army Conference on Applied Statistics, Monterey, CA
- 2005 Experiment Planning Issues in System Reliability. Design and Analysis of Experiments Conference 2005, Santa Fe, NM
- 2005 Bayesian Reliability Analysis for Complex Systems. Joint Statistical Meetings, Minneapolis, MN
- 2005 Reliability Reporting. W88 Annual Assessment Director's Briefing, Los Alamos National Laboratory
- 2005 Reliability Initiatives. W76 Annual Assessment Director's Briefing, Los Alamos National Laboratory
- 2004 Techniques for Test Design in the Context of Staged Acquisition. National Research Council Workshop on Spiral Acquisition, Washington, DC
- 2004 An Overview of Uncertainty in the V&V Context. Foundations 2004: A V&V Workshop, Phoenix, AZ
- 2004 Predicting Operational Reliability with Quantified Confidence from Limited System Data. Putting the "E" in T&E, International Test and Evaluation Association, Baltimore, MD

- 2004 Stockpile Reliability Assessment. ISBA World Meeting, Viña del Mar, Chile
- 2004 Munitions Stockpile Reliability Assessment. Los Alamos National Laboratory High Explosives Working Group, Los Alamos, NM
- 2004 Statistics in Defense and National Security at Los Alamos National Laboratory. Chesapeake Bay Chapter, American Statistical Association, Aberdeen, MD
- 2004 Statistics in Defense and National Security at Los Alamos National Laboratory. Virginia Military Institute, Lexington, VA
- 2003 Opportunities at Los Alamos National Laboratory. Virginia Military Institute, Lexington, VA
- 2003 Assessing Risks for Complex Systems: A Case Study in Missile Defense. Joint Statistical Meetings, San Francisco, CA
- 2002 Combining Information. National Research Council Panel on the Operational Evaluation of the Interim Armored Vehicle
- 2002 Integrating Information Using Belief Functions. Mathematical Methods in Reliability 2002, Trondheim, Norway
- 2002 Statistical Methods for Information Integration. Spring Research Conference on Statistics in Industry and Technology, Ann Arbor, MI
- 2002 Statistics in Defense and National Security. Brigham Young University, Provo, UT
- 2002 Discussion of Information Integration using Belief Functions. Los Alamos Workshop on Novel Approaches to Uncertainty Quantification, Los Alamos, NM
- 2001 Data, Information, and Knowledge Integration for Department of Defense Problems. Joint Statistical Meetings, Atlanta, GA
- 2001 Integrated Analysis of Computer and Physical Experiments. Spring Research Conference on Statistics in Industry and Technology, Roanoke, VA
- 2001 Information Integration Technology. Risk-Based Design Seminar Series, NASA-Langley, Hampton, VA
- 2001 Information Integration Technology. Uncertainty Quantification Working Group, Sandia National Laboratories, Albuquerque, NM
- 2001 Information Integration Technologies. Basic and Applied Simulation Sciences Group, Los Alamos National Laboratory, Los Alamos, NM
- 2001 Quantifying Uncertainty Using Information Integration Technology. Workshop on Uncertainty in Design, Analysis, and Certification of Engineering Systems, Albuquerque, NM
- 2000 Data and Information Integration. Roundtable Luncheon at the Joint Statistical Meetings, Indianapolis, IN
- 2000 Statistical Approaches to DoD Problems at Los Alamos National Laboratory. RAND, Santa Monica, CA
- 2000 Statistical Methods to Validate Command and Control Software. Joint Research Conference on Statistics in Quality, Industry, and Technology, Seattle, WA
- 2000 Data, Knowledge, and Information Integration to Support Decision Making. Military Operations Research Society Education Symposium, McLean, VA
- 1996 A Bayesian Image Model for Automated Feature Identification. Joint Statistical Meetings, Chicago, IL
- 1994 Using Features to Model Prior Structural Information. Spring Research Conference on Statistics in Industry and Technology, Chapel Hill, NC

- 1992 Implementing the Gibbs Sampler on a Parallel Machine: Computationally Intensive Statistics Meets the Hypercube, National Institutes of Health, Bethesda, MD

Contributed

- 2015 Estimating Mission Reliability from Heterogeneous Data. Conference on Applied Statistics in Defense, George Mason University, Fairfax, VA
- 2003 Uncertainty Quantification for Complex Engineered Systems. JANNAF 39th Combustion/27th Airbreathing Propulsion/21st Propulsion Systems Hazards/3rd Modeling and Simulation Joint Subcommittee Meeting, Colorado Springs, CO
- 2003 Information Integration for Stockpile Surveillance. U.S. Army Conference on Applied Statistics, Napa, CA
- 2002 Munitions Stockpile Reliability Assessment. U.S. Army Conference on Applied Statistics, Raleigh, NC
- 2001 Information Integration Technology for Reliability Assessment. 69th MORS Symposium, Annapolis, MD
- 2000 Information Integration: An Integration of Computer and Experimental Data with Expert Opinions. Poster at Threat Reduction Science and Technology Showcase at Los Alamos, Los Alamos, NM
- 2000 Information Integration: An Integration of Computer and Experimental Data with Expert Opinions. Poster at Celebrating Scientific Excellence Through Diversity At LANL, Los Alamos, NM
- 1999 Validating Control System Software. Poster at the Third Annual DOE/MICS Workshop, Los Alamos, NM
- 1999 Statistical Methods to Validate the National Missile Defense Battle Management Command and Control Software. U.S. Army Conference on Applied Statistics, West Point, NY
- 1998 Modeling Error in Probability-of-Kill Tables. U.S. Army Conference on Applied Statistics, Las Cruces, NM
- 1997 Discussion of Practical Inferences for Decision Making in Industrial Applications. Joint Statistical Meetings, Anaheim, CA
- 1996 Statistical Models for Shapes and Deformations. Special contributed paper for Savage Award session. Joint Statistical Meetings, Chicago, IL
- 1995 Models for Shape Deformation. Poster at the Fifth Valencia International Meeting on Bayesian Statistics, Alicante, Spain
- 1994 Priors on Scale-Space Templates. International Symposium on Optics, Imaging, and Instrumentation, San Diego, CA
- 1992 Remarks on the Gibbs Sampler and its Implementation on a Parallel Machine. Joint Statistical Meetings, Boston, MA

CERTIFICATIONS AND LEADERSHIP TRAINING

Project Management Professional (PMP®)

Accredited Professional Statistician (PStat®)

Leadership for a Diverse Campus, North Carolina State University

Task Leader Training, Institute for Defense Analyses

Los Alamos National Laboratory Management Institute: Situational Leadership, Coaching Skills
for Managers, Strategic Project Thinking, Managing Change, Managing Technical
Professionals in Organizations
Los Alamos National Laboratory, Project Management Toolkit